

Techno-economic investigation of an environmentally friendly small-scale solar tracker-based PV/wind/Battery hybrid system for off-grid rural electrification in the mount bamboutos, Cameroon

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ARTICLE INFO

Handling Editor: Mark Howells

Keywords:

Techno-economic investigation
PV
Wind
Battery hybrid system
Solar tracker
Net present cost (NPC)
Levelized cost of Energy (LCOE)
HOMER Pro

ABSTRACT

The present work demonstrates the techno-economic analysis of an environmentally friendly small-scale PV/Wind/Battery hybrid system for off-grid rural electrification in the city of Mbouda. The major contributions of this work are fourfold: (i) PV/Wind/Battery hybrid system is proposed for the very first time in this locality; (ii) six PV tracking techniques namely: horizontal axis (monthly adjustment), horizontal axis (weekly adjustment), horizontal axis (daily adjustment), horizontal axis (continuous adjustment), vertical axis (continuous adjustment) and dual-axis tracker are implemented in the present hybrid system; (iii) the impact of ambient air temperature on the PV power production is evaluated; (iv) the influence of three different hub heights (17 m, 50 m, 150 m) on the wind turbine power output is investigated. The electric load for the basic needs of a community of 152 consumers (households, hospitals, schools, businesses) such as; lighting, radio, television, phone charger, water heater, electric iron, and laptop, is estimated. HOMER Pro is used for the optimization and sensitivity analysis of the proposed hybrid system. The solutions obtained in the present work showed that compared to the PV power generated by fixed solar panels, the dual-axis solar tracker system was the system that offered the maximum power production (about 7.02%–30.73% more power). In addition, compared to the PV penetration of fixed solar panels, the dual solar tracker also offered the maximum PV penetration (about 7.64%–31.50% more PV penetration). It was also concluded that increasing the hub height (from 17 m to 150 m), significantly increased the wind power generation and penetration. Concerning the economic aspect of the work, it was noticed that the dual-axis solar tracking system presented the lowest levelized cost of energy (LCOE) and net present cost (NPC) when compared with the other aforementioned solar tracking systems. The results also indicated that taking into account the ambient air temperature, impacted the PV and wind power productions. The analysis of emissions allowed us to confirm that the system was environmentally friendly since no greenhouse gas was produced.

1. Introduction

The energy deficit is a major concern in various countries in the world in general and in developing countries such as Cameroon in particular. To remedy this deficit, renewable energy technologies were adopted by several governments. Concerning the solar energy, it has a major problem related to its intermittency and dilute nature. This issue can be corrected by the association of a solar tracker in the architecture of the energy system. A solar tracker (ST) is an equipment that orients a payload (solar panel, parabolic trough, fresnel reflectors, lenses) toward the sun. It was widely used in the literature in the design of energy

systems for the improvement of their efficiencies. Indeed, an electronic sun finder and ST consisting of two photodiodes separated from each other by an opaque body, in an enclosed rectangular structure was proposed by Salawu and Oduyemi [1]. It was demonstrated that the system reasonably tracked the sun. Al-Naima designed and constructed a ST. Its performance was evaluated on a PV array mounted on it [2]. Helwa et al. evaluated the maximum collectable solar energy by different ST and showed that their electricity productions were improved when compared with that of a stationary system [3]. They concluded that the gains were improved with the two-axis ST and decreased gradually from the vertical-axis ST to the tilted-axis ST. The

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<https://doi.org/10.1016/j.esr.2023.101107>

Received 26 January 2023; Received in revised form 8 May 2023; Accepted 19 May 2023

Available online 22 May 2023

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same authors computed the solar energy captured by different ST (fixed systems facing south and tilted 40° , a vertical-axis ST, a 6° tilted-axis ST, and a two-axis ST) [4]. The precisions obtained annually by the investigators were 5.36%, 9.07%, 7.92%, and 5.98% respectively. Clifford and Eastwood designed a passive ST for an utilization in equatorial regions around the world [5]. It was shown an excellent agreement between the experiment and the computer model (an increase in efficiency of up to 23% over fixed PV). Piao et al. developed a 150 W ST which minimized the solar cosign loss of the PV array [6]. Experiments presented 10,159 and 12,360 Wh of electrical energy generation for fixed and ST respectively. Karimov et al. proposed a PV tracking system with a tracking angle of 120° . This system was found to be very economical and cost-effective [7]. A genetic algorithm using ST approach was developed in order to improve the PV performance [8]. The proposed algorithm showed a capability to adapt to different environmental conditions and found the maximum power production by a suitable orientation of the PV panels. Kelly and Gibson used a ST to improve the PV output for cloudy conditions [9]. They showed that the horizontal tilt ST captured the most solar energy. Barsoum designed and constructed a prototype of ST with two degrees of freedom. It was based on a PIC16F84A microcontroller [10]. The results allowed us to conclude that single-axis freedom ST increased energy output by approximately 20%, whereas double-axis freedom ST system increased the output by more than 40%. A controller ST with minimal cost was developed [11]. A low-cost dual-axis ST using an adaptive solar sensor was proposed by Arge-seanu et al. [12]. Tina and Gagliano presented a probabilistic model applied to a wind-PV hybrid system equipped with either a one-axis or a two-axis ST [13]. They concluded that two-axis ST improved the system efficiency when compared with the one-axis ST. Chin et al. designed, modeled, and tested a standalone single-axis active ST using Matlab simulink [14]. The system provided the flexibility to accommodate different weather conditions and user preferences. An intelligent ST based on an embedded LPC2131 microcontroller was implemented [15]. It improved the PV conversion efficiency by tracking the movement of the sun through a two-axis stepping motor rotation. An optimized use of ST and wind energy using HOMER energy was proposed by Bhattacharya et al. [16]. They demonstrated that the efficiency of the system increased by around 13.9%. Zolkapli et al. proposed a high-efficiency dual-axis ST using Arduino [17]. The resulting system had the capability to capture the maximum sunlight. Elagib and Osman described the design and development of a microcontroller-based ST, based on solar maps [18]. The system offered different advantages: high stability; the control algorithm used had no searching phase; it calculated the position of the sun and moved the motors towards the sun immediately. León et al. presented a semi-passive ST concentrator whose configuration required minimal mechanical effort [19]. An optimum fixed array's tilt of 49.054° was found to maximize the effective Fresnel lens area by reducing blocking and shading, both caused by the position of the array above the lens. A multipurpose dual-axis ST with two tracking strategies (normal tracking strategy and daily adjustment strategy) was applied on a PV [20]. The results suggested that the average energy efficiency of the normal tracking PV, compared with the fixed PV, was more than 23.6%; and the average energy efficiency of the daily adjusted PV was more than 31.8%. Stamatescu et al. presented a ST to control a PV panel movement in order to improve the conversion efficiency of the system [21]. They showed that the developed algorithm reduced the costs of the tracking method and made it a cost-effective technology. Sidek et al. designed and developed a portable dual-axis ST that focused on portable and mobility purposes [22]. The technique was based on astronomical equations and was capable to calculate and generate the sun's trajectory path through the year. Khadidja et al. optimized a ST for PV power plants in the Saharian region [23]. The authors concluded that it was possible to gain a significant amount of energy when mounting PV systems on trackers. This gain depended on the location. Vermaak carried out a techno-economic analysis of ST in South Africa [24]. He realized that for the single as well as for the two-axis system, there was

an additional gain of energy of about 33% and 37% respectively compared to the static system. A simple ST based on light dependent resistors was constructed, tested and optimized by Bentaher et al. [25]. A methodology of an automatic ST was proposed by means of two sensors and image processing simultaneously [26]. It was concluded that the proposed system was more accurate and efficient than the conventional ST. Zhang et al. proposed an automatic ST to produce power for lighting greenhouses [27]. The authors concluded that compared to the power generated by fixed solar panels, the ST generated about 20%–25% more power. Das et al. performed a design and an experimental execution of a microcontroller-based smart dual-axis automatic ST [28]. The authors indicated that the power output of the PV system, using the proposed tracking system, was increased up to 19.73% when compared to the traditional fixed PV panel. Fathabadi developed a high accurate sensorless dual-axis ST controlled by maximum power point tracking unit of PV systems [29]. He obtained an increase in the energy efficiency ranging between 28.8% and 43.6%, depending on the seasons. Huang et al. proposed a two-axis ST using fuzzy logic as intelligent quality policy [30]. The results suggested that the approach reduced the number of starting motors and resulted in smaller energy loss in cloudy, cloud mask, or unstable weather conditions. Parthipan et al. designed a one-axis three-positions ST for paraboloidal dish solar collector [31]. They showed that the system was able to track and follow the sunlight intensity in order to get maximum solar energy in the form of heat irrespective of the motor speed. Sharaf Eldin et al. studied the feasibility of ST for PV panels in hot and cold regions [32]. The investigators realized that the gain in electrical energy from tracking the sun was about 39% in case of a cold city, while the gain in energy did not exceed 8% in case of a hot city due to overheating of the PV panels. Burhan et al. proposed a compact but highly accurate and cheap two-axis ST, designed for concentrated PV system [33]. The system was designed for tracking accuracy requirement of 0.3. Wei et al. developed an innovative method to track the sun using an image sensor [34]. The results indicated that the system successfully captured the real sun center in most weather conditions. Mi et al. constructed an open-loop ST strategy for high concentrating PV systems using variable tracking frequency [35]. They reduced the total moving frequency of the ST by implementing the proposed strategy. Hong et al. performed a preliminary study on the two-axis hybrid ST for the smart PV blind [36]. The slope of PV panel tracked the sun from 0° to 90° , while the azimuth of PV panel tracked the sun from -9° to 9° due to the limitation of the rotating angle of the vertical axis. Racharlan and Rajan reviewed different types of ST and confirmed that the azimuth and altitude dual-axis ST was more efficient compared to other tracking systems [37]. They concluded that in terms of cost and flexibility, one-axis ST was more feasible than dual-axis ST. Sidek et al. studied an automated positioning dual-axis ST with precision elevation and azimuth angle control [38]. It was found that the system was able to produce 26.9% and 12.8% higher power than fixed-tilted PV system on a clear and heavy overcast conditions respectively. A ST of a nonelectric power source by using a metal hydride actuator was developed by Obara et al. [39]. Sumathi et al. reported a review of various methods of ST with gains in energy due to tracking and different maximum power point tracking algorithms [40]. It was observed that the increase in gain due to active tracking was approximately 30% with respect to the fixed system. Fernández-Ahumada et al. presented a mathematical study of the movement of ST based on rational models [41]. A dual-axis ST that worked based on processing images of a bar shadow was developed [42]. It was indicated that the ST followed the sun with an accuracy of about $\pm 2^\circ$ and maintained the panel perpendicular to the irradiation direction. Carballo et al. built a prototype of low-cost ST based on an open source hardware for educational purposes [43]. Hafez et al. conducted a review on ST and their potentials in solar energy applications [44]. Safan et al. presented a design, modeling and simulation of a two-axis ST with a hybrid controlled tracking system [45]. They obtained a sun tracking pointing error of $\pm 0.18^\circ$ and a typical tracking

accuracy of $\pm 0.12^\circ$. Krishna et al. evaluated the performance of static and tracking-type PV systems [46]. They concluded that the tracking system output increased from 15 to 20% than the fixed one. Nsenyumva et al. reviewed the most commonly used ST and identified the systems that offered benefits such as greater efficiency, greater tracking accuracy, easy installation and cost effectiveness [47]. Their study revealed that dual-axis ST was the most effective and generally gave more than 40% gain in energy return compared to fixed PV panels. A ST was also coupled to a solar water heater and the thermal efficiency of the system was analyzed [48]. It was noticed a 40% gain of overall stored thermal energy for ST system. Batayneh et al. investigated a single-axis discrete ST for reduced actuations and maximum energy collection [49]. The experiments showed that tracking the sun in only three optimized angle movements yielded around 91–94% of solar energy collection as a continuous single-axis ST that moved three times an hour based on solar calculations. A comparative analysis of the performance of three types of ST (fixed, single and dual-axis) in the context of Bangladesh was carried out by Fahad et al. [50]. It was observed that the difference in the total incident energy in one whole year between the single and dual-axis ST was about 3.96% without considering the effect of cloud which further reduced to 3.44% when the effect of cloud cover was taken into account. Fazlizan et al. analyzed a dual-axis ST with smart algorithms called maximum light detection applied in the solar energy generation system

[51]. The comparative results showed that the system with maximum light detection tracker improved the energy generation significantly for all sky conditions. Mpodi et al. investigated a dual-axis ST [52]. It was found that the efficiency of dual-axis ST was about 35–43%. A concentrative PV solar system was constructed and proposed [53]. The proposed system showed 16.58% higher power productivity, compared to a pinhole system and other existing methods. A review on various technologies of ST to determine the best PV panel orientation was presented [54]. The conclusion made by the authors was that the dual-axis rotation was the most favorable ST. Das and Akpinar investigated the impacts of ST on the performance of solar air dryer [55]. The results indicated that the use of a ST improved significantly the solar air collector thermal efficiency. Maliani et al. developed a solar still coupled with a dual-axis ST [56]. The yield of their proposed solar still was about 3.76 L/day. A design and implementation of efficient one- and dual-axis ST control systems were proposed [57]. The model proposed by the authors provided better results with five membership functions. Zhu et al. designed and analyzed the performance of a ST with a novel single-axis tracking structure to maximize the energy collection [58]. Das and Akpinar determined the thermal and drying performances of the solar air dryer with ST [59]. The solar air collector thermal efficiencies were about 75.7% and 51.7% for moving and fixed solar air collector dryer respectively. ElGamal et al. developed an integrated ST for enhancing



Fig. 1. Map of the site of study [81].

the performance of solar air heaters in drying apple slices [60]. The results suggested that the thermal efficiency of the solar air heater incorporated with a tracking unit was significantly improved by about 45% compared with the conventional fixed heaters at all tested air flow rates. A dual-axis ST with mirror reflection using arduino uno microcontroller as the control unit was proposed [61]. The conclusions drawn by the investigators were that the dual-axis ST with mirror reflection provided better efficiency and improved the performance of the solar panel. Manosroi et al. studied the performance improvement of a two-axis ST by using flat-mirror reflectors [62]. Awasthi et al. depicted the ST technologies in solar PV system [63]. They concluded that dual-axis ST was more efficient and advantageous than its single-axis and fixed counterparts. Rad et al. studied the techno-economic and environmental features of different ST for residential PV installations [64]. A systematic review of the control algorithms implemented in active ST was presented by Fuentes-Morales [65]. Munanga et al. developed an intelligent single-axis ST device [66]. The resulting system showed a 25% increase in efficiency. Barbón et al. designed and constructed a ST for a small-scale linear Fresnel reflector with three movements [67]. Taheri et al. conducted an experimental research for the improvement of the electrical and thermal output power of a PV thermal module equipped with an automatic dual-axis ST [68]. It was noticed an increase of the overall energy/exergy of the system when it was coupled with a ST. A standalone ST was successfully synchronized using proteus 7.6 ISIS [69]. The results demonstrated that the maximum electrical current drawn by the ST was less than 0.5 mA. A PV system moving both in east-west horizontally and in a north-south vertically directions was designed [70]. The proposed design produced approximately 24.7% more power than the 45° fixed-tilted system. A ST with new hybrid control in energy production optimization from PV conversion for polish climatic conditions was suggested [71]. The increases in the generated electricity ranging between 35.6% and 44.7% were observed for local climatic conditions. A dual-axis ST based on a light-to-frequency converter using a microcontroller was constructed by Ghassoul [72]. An electromechanical ST for a parabolic dish with a central processing unit water heater was developed [73]. Baouche et al. described a methodology for the simulation and the design of a ST using the advantages that the orientation and efficiency of the PV panel offered due to the latitude and the number of hours of sunshine [74]. Their results provided a good platform for engineering technology researchers and students to study the design theory of a ST. A two-axis ST was developed and employed for the improvement of the energy output of a concentrating solar system that produced hot water and light [75]. The experimentation showed that the maximum efficiency of the hybrid solar lighting/thermal system controlled with a dual-axis ST was 32.2%. Wu et al. proposed an enhanced ST to solve the issues of low sunlight incident angles and manual orientation/elevation calibrations [76]. Tchakounte et al. constructed and compared two approaches of ST [77]. They suggested that the manual method could increase the performance of a PV module up to 21% compared to the fixed module. A low-power

consumption ST was implemented for driving a crystalline silicon PV module under variable weather conditions [78]. The solutions showed that this ST consumed less than 1% of the increased energy. Taking into account the ST consumption, the energy gain could attain up to 25% under the same weather conditions compared to a fixed PV system. An enhanced ST to solve the issues of low sunlight incident angles and manual orientation/elevation calibrations was proposed by Wu et al. [79]. Nieto et al. proposed an economic study of flywheel energy storage systems integration with a grid connected intermittent power plant, for power quality purposes [97]. Indeed, the economic performance of a 99 MW wind farm interconnected in a generic electrical power grid was analyzed. Three different power capacities for flywheel energy storage systems (25 MW, 30 MW and 40 MW) were considered. They authors concluded that all the scenarios were feasible since the net present value (NPV) was positive and higher than £1 M. In addition, it was noticed that the internal rate of return (IRR) was higher than the discount rate of the project. In the same vein, Nieto et al. explored the impact of a wind power plant on a power grid and integrated energy storage systems of different capacities to the power grid in order to evaluate the improvements in the power quality [98]. They showed that with the integration of the energy storage systems, the backup power being required from the slack generator was reduced to 16.82% with the application of a 40 MW, 19.72% with a 30 MW and 21.17% with a 25 MW energy storage systems. They also demonstrated that a scenario with wind power plant output of 52.5 MW or above, the 40 MW energy storage system could support the injection of active power to the bus 5 for being above 92.5 MW. Ghaithi et al. developed the technical and economic viability of hybrid energy system in the Masirah Island power system in Oman [99]. They used the HOMER software to optimize the proposed hybrid system and DiGSILENT software for the modeling and simulation of the integration of the proposed hybrid system to the existing network. The investigators showed that the proposed architecture made up of PV, diesel, and wind turbine was the most economically feasible system since it provided the lowest system net present cost, operating cost and cost of energy, while it also improved the voltage profile at the point of connection. Seritan et al. performed an evaluation of the performance of PV panels containing cells with different bus bars configurations in partial shading conditions [100]. They analyzed seven types of poly-Si PV cells connected in series. They proved that the performance of a cell was given by a perfect balance between effective exposed area and internal resistance. Sambhi et al. developed a technical and economic analysis of PV/Diesel hybrid system using five different battery storage technologies (lead-acid, lithium-ion, vanadium flow, zinc bromide and nickel-iron) [101]. Their optimal system was composed of a PV array of 200 kW, a diesel generator of 82 kW, a battery storage system with a nominal capacity of 2000 kWh and a converter of 68 kW. The initial cost, operating cost, net present cost and cost of energy were respectively USD163,445; USD534/year; USD170,348; and USD0.090/kWh. Sambhi et al. performed a technical, economic and environmental analysis for a hybrid energy system generating a part load at SRM IST at the Delhi-NCR campus, India [102]. The proposed hybrid system was designed to meet the electric load demand of 400 kWh/day with a peak load of 74.27 kW and hydrogen load demand of 10 kg/day with a peak demand of 1.86 kg/h. The analysis was performed for both on-grid and off-grid conditions. The cost of energy was \$0.408/kWh, while the cost of hydrogen was \$16.6 per kg.

A careful analysis of this literature review clearly indicated that the ST was largely utilized to improve the power output of PV systems in various meteorological conditions in the world. Unfortunately, it is worth mentioning that these systems are not popularized in some third world countries (such as Cameroon), although there are many off-grid electrification projects based on PV modules. The novelty of the present work is fourfold: (i) PV/Wind/Battery hybrid system is proposed for the very first time in this locality; (ii) six PV tracking techniques namely: horizontal axis (monthly adjustment), horizontal axis (weekly adjustment), horizontal axis (daily adjustment), horizontal axis (continuous

Table 1
Approximation of the daily load profile.

Appliance	Power (W)	Quantity	Number of hours/day	Total power (W)	Energy (Wh)/day
Phone charger	5	3	6	15	90
Radio	15	1	4	15	60
Lighting bulbs	15	4	4	60	240
Water heater	1000	1	0.125	1000	125
TV	36	1	4	36	144
Electric iron	1000	1	0.125	1000	125
Laptop	15	1	3	15	45
Total					829

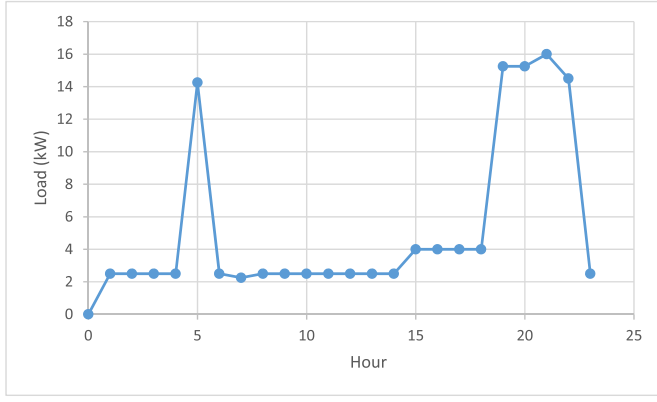


Fig. 2. Approximation of the daily load profile.

adjustment), vertical axis (continuous adjustment) and dual axis tracker are implemented in the present hybrid system; (iii) the impact of ambient air temperature on the PV power production is evaluated; (iv) the influence of three different hub heights (17 m, 50 m, and 150 m) on the wind turbine power output is investigated. The rest of the paper is structured as follows: Section 2 presents the methodology, section 3 shows the results and discussions and concluding remarks are presented in section 4.

2. Methodology

HOMER Pro software was developed by the National Renewable Energy Laboratory and was used for optimizing the design and economic performance of the proposed PV/Wind/Battery hybrid system. This platform examines all possible combinations of system types in a single run, then sorts the systems based on the chosen optimization variable [80]. It proposes various energy components, namely, photovoltaic, wind turbines, hydro, batteries, diesel and other fuel generators, electrolysis units, and fuel cells. HOMER Pro evaluates suitable options taking into account cost and availability of energy resources. It requires initial information including energy resources, economical and technical constraints, energy storage requirements and system control strategies. The main advantage of HOMER is that it combines reliability and minimum possible cost to propose configurations of the system. Its disadvantage is that the optimization procedure is not clearly described for commercial reasons. In addition, it does not consider the bus voltage variation and users cannot change some of its design parameters [103].

2.1. Study location

The site of study is the mount bamboutos (latitude: $5^{\circ}40'0''$ North

and longitude: $10^{\circ}3'0''$ East) located in the city of Mbouda in the West region of Cameroon. The average annual solar radiation is $4.86 \text{ kWh/m}^2/\text{year}$. Wind speeds reach 3.3 m/s in the site. A map of the site of study is shown in Fig. 1 [81].

2.2. Approximation of the load profile

The approximation of the load profile was made possible from a survey in the city of Mbouda where a community of 152 consumers (households, hospitals, schools, businesses) were enrolled. They used electricity for their basic needs such as, for lighting, radio, television, phone charger, water heater, electric iron, and laptop. It was observed a peak load of 105.27 W in the evening. The daily load profile is estimated in Table 1. Fig. 2 shows the daily load profile plot.

2.3. Resources of solar and wind energy

2.3.1. Wind energy resources

The electricity output produced by a wind turbine is evaluated using the characteristic curve of the manufacturer and the wind speed at hub height defined by Ref. [82]:

$$v = v_0 \left(\frac{h}{h_0} \right)^{\gamma} \quad (1)$$

with v , v_0 and γ the wind speeds at the hub heights h , h_0 and the ground surface friction coefficient. Practically, $\gamma = \frac{1}{7}$ [82]. Concerning the power output, it is given by Ref. [82]:

$$P_{out} = \begin{cases} 0 & \forall v < v_{cut-in} \\ a_{11}v^n \dots a_{12}v^2 + a_{13}v + a_{14} & \forall v_{cut-in} \leq v < v_1 \\ a_{21}v^n \dots a_{22}v^2 + a_{23}v + a_{24} & \forall v_1 \leq v < v_2 \\ a_{31}v^n \dots a_{32}v^2 + a_{33}v + a_{34} & \forall v_2 \leq v < v_{cut-off} \\ a_{31}v^n \dots a_{32}v^2 + a_{33}v + a_{34} & \forall v > v_{cut-off} \end{cases} \quad (2)$$

where v_{cut-in} , $v_{cut-off}$, v , v_1 , v_2 and P_{out} are the cut-in speed of the wind turbine, cut-off speed of the wind turbine, wind speed, intermediate wind speed levels, and the generated power of the wind turbine. The monthly average wind speed profile at 10 m height at the mount Bamboutos is illustrated in Fig. 3. These data were collected from the NASA surface meteorology and solar energy database. It is observed in Fig. 3, the highest values of wind speeds (about 3.3 m/s) during the rainy season (July and August). The lowest value occurred in the month of May (dry season).

2.3.2. Solar energy resources

The estimation of monthly average solar radiation can be achieved through the following equations [83]:

$$\left\{ \begin{array}{l} S = S_0 \left(a + b \frac{n}{N} \right) \\ S_0 = \frac{24 \cdot 3600 \cdot G_{sc}}{\pi} \left(1 + 0.033 \cdot \cos \left(\frac{360}{365} n_d \right) \right) \left(\cos \varphi \cos \delta \sin \omega_s + \frac{\pi \omega_s}{180} \sin \varphi \sin \delta \right) \\ \delta = 23.45 \sin \left(360 \left(\frac{248 + n_d}{365} \right) \right) \\ \omega_s = \cos^{-1} (-\tan \varphi \tan \delta) \\ N = \frac{2\omega_s}{15} \end{array} \right. \quad (3)$$

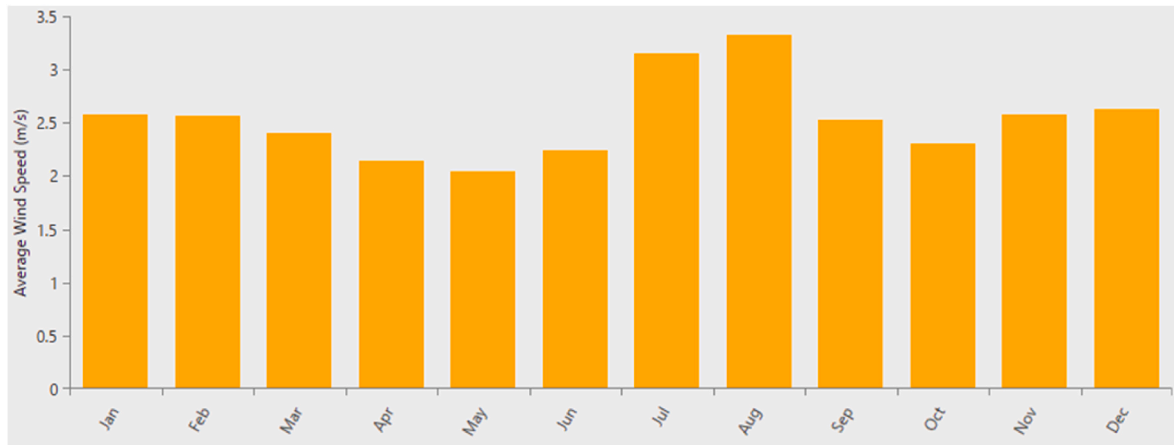


Fig. 3. Monthly average wind speed profile at 10 m height at the mount Bamboutos.

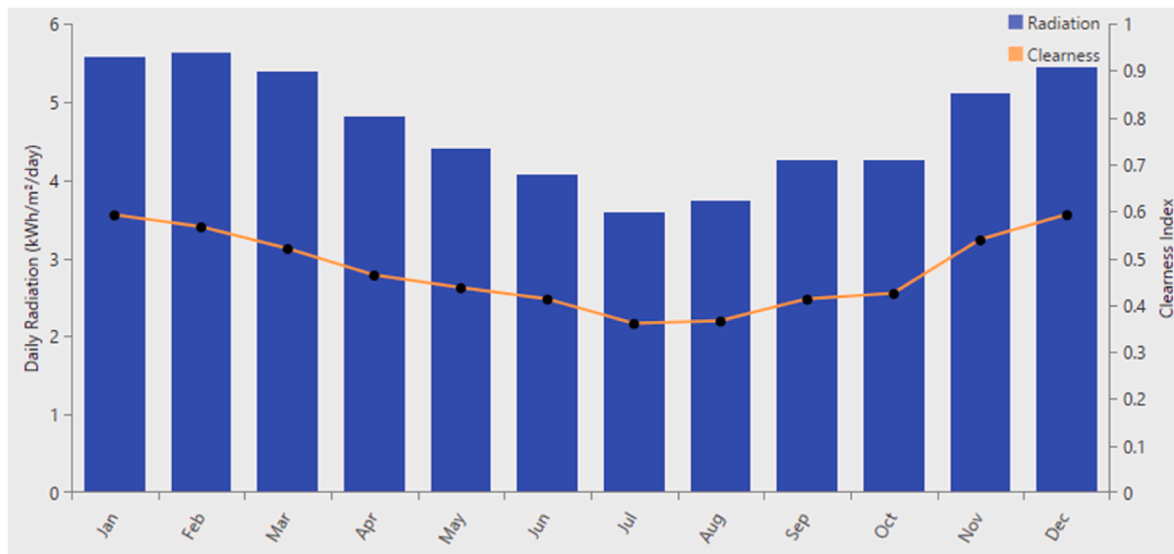


Fig. 4. Monthly average solar radiation at the mount Bamboutos.

With S , S_0 , N , n , a and b the monthly average daily radiation on horizontal surface (MJ/m^2), monthly average daily extraterrestrial radiation on a horizontal surface (MJ/m^2), maximum possible daily hours of bright sunshine, monthly average daily number of hours of bright sunshine and regression coefficients respectively. The average values a and b are: $a = 0.33$ and $b = 0.43$. n_d , G_{sc} , φ , δ and ω_s are the day number starting from the first of January, solar constant ($1367 \text{ W}/\text{m}^2$), latitude of the location, declination angle and Sunset hour angle respectively [83]. The data of solar radiation at the mount Bamboutos were obtained from the NASA website [84] (see Fig. 4). The annual average solar radiation and the average clearness index were $4.69 \text{ kWh}/\text{m}^2/\text{day}$ and 0.473 respectively at the site of study. The highest value of solar radiation is observed in the month of February and the lowest value in the month of July.

2.3.3. Solar tracking techniques based on the axis

A solar tracking technique is a method to improve the efficiency of a PV module. Solar trackers move the payload towards the sun throughout the day. Various solar tracking methods, among which those based on the axis exist in the literature [37].

2.3.3.1. Single-axis tracker. It represents a technology that adjusts the position of a PV module along an axis (one degree of freedom that acts as

an axis of rotation) to follow the sun's changing position throughout the passing days and years [37]. There exists several single-axis trackers in the literature, namely: horizontal (with horizontal axis of rotation with respect to the ground), vertical (with vertical axis of rotation with respect to the ground), tilted (with axes of rotation between horizontal and vertical) and polar aligned single-axis (for which the tilted single axis is aligned to the polar star) trackers [37]. Fig. 5 presents the fixed-tilt solar panel, horizontal and vertical single axis trackers.

2.3.3.2. Dual-axis tracker. Dual-axis trackers continually face the sun because they can move in two different directions. Many dual-axis trackers are found in the literature: Tip-tilt (for which the PV module is mounted on the top of a pole), and azimuth-altitude with its primary axis (azimuth axis) vertical to the ground and the secondary axis (elevation axis) normal to the primary axis [37]. Fig. 6 shows the dual-axis tracker.

In the present proposed work, six PV tracking techniques namely: horizontal axis (monthly adjustment), horizontal axis (weekly adjustment), horizontal axis (daily adjustment), horizontal axis (continuous adjustment), vertical axis (continuous adjustment) and dual-axis trackers are implemented in the present hybrid system through HOMER Pro.

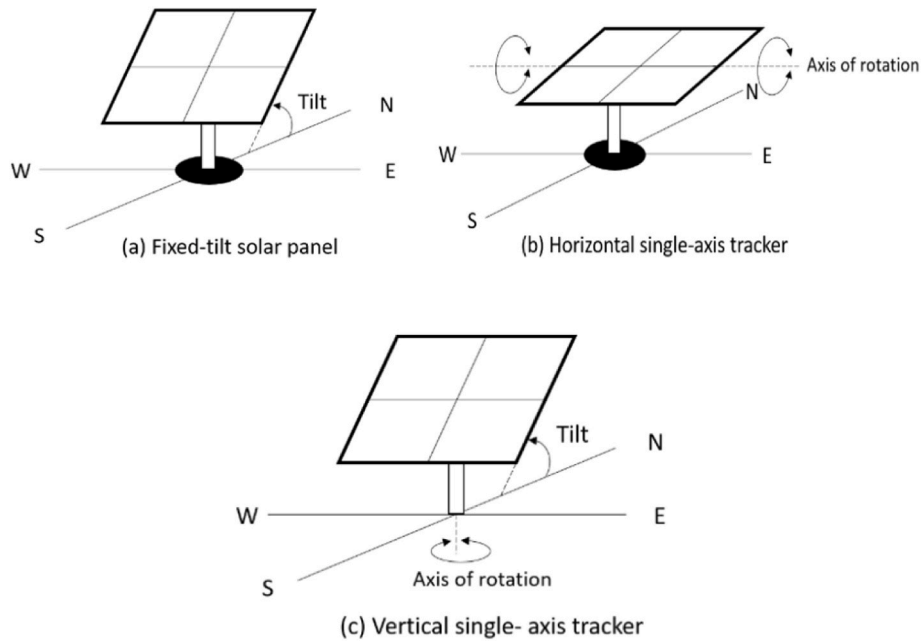


Fig. 5. Fixed-tilt solar panel, horizontal and vertical single-axis trackers [93].

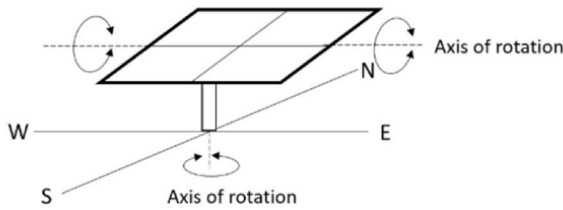


Fig. 6. Dual-axis tracker [93].

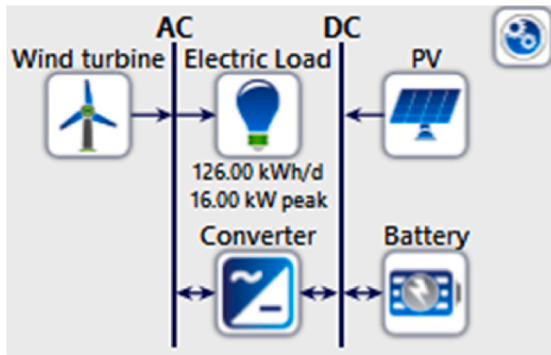


Fig. 7. Hybrid system model implemented in HOMER Pro.

2.4. Hybrid system components modeling using HOMER pro

A hybrid system made up of PV arrays, batteries, converter and wind turbine was modeled by using HOMER Pro. Six PV tracking techniques namely: horizontal axis (monthly adjustment), horizontal axis (weekly adjustment), horizontal axis (daily adjustment), horizontal axis (continuous adjustment), vertical axis (continuous adjustment) and dual-axis trackers were implemented in this work in the software. The impact of ambient air temperature on the PV power production was evaluated and three different hub heights (17 m, 50 m, and 150 m) of the wind turbine were considered. Fig. 7 shows the hybrid system model implemented in HOMER Pro.

Table 2

Details of the technical parameters for the PV arrays [86].

Model	BP350
Maximum power (W)	50
Nominal voltage (V)	12
Maximum power voltage (V)	17.5
Maximum power current (A)	2.9
Open-circuit voltage (V)	21.8
Short-circuit current (A)	3.17
Surface (m ²)	0.451
Capital cost (\$)	148.5
Replacement cost (\$)	133.65
Operation and maintenance cost (\$)	14.85

2.4.1. PV modules

The principle of operation of photovoltaic modules is based on the photovoltaic effect. Indeed, the photovoltaic effect is obtained by absorption of photons in a semiconductor material which generates an electric voltage. Photovoltaic cells emit direct current from solar radiation, which can be used to power a device or recharge a battery. The power output generation of a PV module can be computed using [85]:

$$P_{output} = \eta_r \eta_{PC} (1 - \beta(T_a - T_{ref} + ((T_{op} - 20) / 800)G_t)) A_{PV} G_t \quad (4)$$

With η_{PV} , η_r , η_{PC} , A_{PV} , T_a , T_{ref} , T_{op} and G_t the photovoltaic efficiency, reference module efficiency, power conditioning efficiency, area of photovoltaic generator in m², ambient temperature in °C, cell reference temperature in °C, nominal operating temperature of the cell in °C and solar radiation in W/m². β is the temperature coefficient which takes values between 0.004 and 0.006 per 1 °C for silicon cells [85]. The

Table 3

Technical details for the converter [89].

Item	Value
Capital cost (\$)	985
Replacement cost (\$)	887
Operation and maintenance cost (\$)	98
Lifetime (years)	20
Efficiency (%)	95
Relative capacity	80%

Table 4
Technical details for the batteries [91].

Item	Value
Nominal capacity (Ah)	253
Voltage (V)	12
Minimum charge (%)	20
Capital cost (\$)	465
Replacement cost (\$)	418.5
Operation and maintenance cost (\$)	46.5

Table 5
Wind turbine characteristics [92].

Model	BWCXL 1000
Power rate (W)	1000
Nominal voltage (V)	24
V_{ci} (m/s)	2.5
Capital cost (\$)	2500
Replacement cost (\$)	2250
Operation and maintenance cost (\$)	250

details of PV arrays used in the design are shown in Table 2 [86].

2.4.2. Converter

A power converter is an electrical circuit that changes the electrical energy from one form into the desired form optimized for the specific load. A converter may do one or more functions and give an output that differs from the input. It is used to increase or decrease the magnitude of the input voltage, invert polarity, or produce several output voltages of either the same polarity with the input, different polarities, or mixed polarities [87]. The power rating of the converters can be obtained through the following expression [88]:

$$P_{inv} = \frac{P_{peak}}{\eta_{inv}} \quad (5)$$

with P_{peak} the peak of the power consumption load and η_{inv} the inverter efficiency. The technical details for the converter are presented in Table 3 [89].

2.4.3. Batteries

The computation of charging state in batteries in each time step reads [90]:

$$\begin{cases} C_{Bcharg}(t) = C_{Bcharg}(t-1)(1-\sigma) + \left(P_T(t) - \frac{P_L(t)}{\eta_{conv}} \right) \eta_{Batt} \\ C_{Bdischarg}(t) = C_{Bdischarg}(t-1)(1-\sigma) + \left(\frac{P_L(t)}{\eta_{conv}} - P_T(t) \right) \eta_{Batt} \end{cases} \quad (6)$$

with $C_{Bcharg}(t)$ and $C_{Bdischarg}(t)$ the capacities of the battery bank during the charging mode and the discharging mode respectively. $P_T(t)$ and $P_L(t)$ are the total output power generated by the system and the load demand at the corresponding hour respectively. η_{Batt} and η_{conv} are the battery efficiency and the conversion efficiency.

The Concorde (PVX-2120L) batteries were used in the system architecture. The technical details of the batteries are shown in Table 4 [91].

2.4.4. Wind turbine

The characteristics of the wind turbine used in the design are specified in Table 5.

2.5. Economic assessment criteria

The economic comparison between the different architectures is based on Net Present Cost (NPC) and Levelized Cost of Energy (LCOE).

They are defined as [94]:

$$\left\{ \begin{aligned} NPC &= \frac{C_{ann,tot}}{CRF(d,N)} \\ C_{ann,tot} &= C_{ann,cap} + C_{ann,rep} + C_{ann,O\&M} + C_{ann,fuel} - R_{ann,sav} \\ CRF &= \frac{d(1+d)^N}{(1+d)^N - 1} \\ d &= \frac{i-f}{1+f} \\ COE &= \frac{C_{ann,tot} - C_{boiler} \bullet H_{thermal}}{E_{served}} \end{aligned} \right. \quad (7)$$

The quantities $C_{ann,tot}$, $C_{ann,cap}$, $C_{ann,rep}$, $C_{ann,O\&M}$, $C_{ann,fuel}$, $R_{ann,sav}$, CRF , d , N , i , and f represent total annualized cost of the system (\$/year), annualized capital (\$/year), replacement (\$/year), maintenance costs of all components of the system (\$/year), annualized cost of fuels used to feed the power generators (\$/year), annualized total savage value of all system components (\$/year), capital recovery factor which is a function of annual real interest rate (8%), discount rate, project lifespan, nominal interest rate and annual inflation respectively. COE , C_{boiler} , $H_{thermal}$ and E_{served} are levelized cost of energy, boiler marginal cost (\$/kWh), total thermal load served (kWh/year) and the total electric load served (kWh/year). By considering that there is no thermal load served in the system, $H_{thermal} = 0$. In this paper, the discount and inflation rates of 5% and 3% were considered respectively [94]. In the present work, two dispatch strategies were implemented: load following (LF) and cycle charging (CC) [95].

2.5.1. Analysis of sensitivity

The parameters used for sensitivity analysis were: the solar scaled annual average irradiation, nominal discount and expected inflation rates of 8% and 3%, project lifetime of 15 years, and maximum capacity shortage of 10. The sensitivity variables and their values are shown in Table 6.

3. Results and discussions

In this section, the results based on the simulation of six PV tracking techniques namely: horizontal axis (monthly adjustment), horizontal axis (weekly adjustment), horizontal axis (daily adjustment), horizontal axis (continuous adjustment), vertical axis (continuous adjustment) and dual-axis trackers were examined in order to choose the best alternative technical and economic properties of the system. In addition, the impacts of ambient air temperature and hub heights on the wind turbine and PV power productions were analyzed. The operation statue of the optimal design with the best option of PV tracking technology was illustrated. For simplification purposes, the different solar tracking techniques were denoted by.

- Horizontal axis, monthly adjustment tracker (HAMA);
- Horizontal axis, weekly adjustment tracker (HAWA);
- Horizontal axis, daily adjustment tracker (HADA);
- Horizontal axis, continuous adjustment tracker (HACA);
- Vertical axis, continuous adjustment tracker (VACA);
- Dual-axis tracker (DAT).

Table 6
Sensitivity variables and their values.

Sensitivity variable	Values
Nominal discount rate (%)	{8; 9.20; 10.40; 11.60; 12.80; 14.00}
Project lifetime (years)	{15; 17; 19; 21; 23; 25}

Table 7

Optimization results for different PV tracking techniques for a hub height of 17 m without taking into consideration the ambient air temperature effects.

Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	52.3	48.9	48.7	48.7	48.7	48.9	44.5
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	18.6	18.0	18.2	18.2	18.2	17.5	20.7
	Batteries	kWh	703	732	735	735	735	739	824
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	1.08	1.03	1.03	1.03	1.03	1.03	0.997
	NPC	US\$	418,385	400,497	400,079	400,079	400,079	400,429	386,514
Power production	PV array	kWh/year	72,230	70,797	70,748	70,758	71,496	70,963	77,780
	Wind Turbine	kWh/year	6580	6580	6580	6580	6580	6580	6580
	Total	kWh/year	78,810	77,377	77,328	77,338	78,076	77,543	84,360
	Unmet load	kWh/year	34.4	31.3	35.6	35.7	34.3	14.8	22.3
	Excess electricity	kWh/year	22,094	20,679	20,636	20,646	21,376	20,835	27,615
	PV penetration	%	157	154	154	154	155	154	169
	Wind penetration	%	14.3	14.3	14.3	14.3	14.3	14.3	14.3
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

Table 8

Optimization results for different PV tracking techniques for a hub height of 17 m with the consideration of the ambient air temperature effects.

Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	52.7	49.0	48.9	48.9	48.9	48.9	44.5
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	19.0	18.6	18.0	18.0	18.0	17.5	22.4
	Batteries	kWh	687	746	732	732	732	739	812
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	1.08	1.04	1.03	1.03	1.03	1.03	1.00
	NPC	US\$	420,222	402,980	400,497	400,497	400,497	400,429	389,047
Power production	PV array	kWh/year	72,688	71,035	70,999	71,009	71,749	70,963	77,792
	Wind Turbine	kWh/year	6462	6462	6462	6462	6462	6462	6462
	Total	kWh/year	79,149	77,497	77,460	77,471	78,211	77,424	84,254
	Unmet load	kWh/year	34.8	19.6	35.6	35.6	34.2	18.2	32
	Excess electricity	kWh/year	22,426	20,776	20,760	20,771	21,503	20,712	27,513
	PV penetration	%	158	154	154	154	156	154	169
	Wind penetration	%	14	14	14	14	14	14	14
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

Table 9

Optimization results for different PV tracking techniques for a hub height of 50 m without taking into consideration the ambient air temperature effects.

Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	46.6	42.5	42.5	42.5	42.5	42.2	40.2
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	18.5	19.1	19.1	19.1	19.1	18.7	18.3
	Batteries	kWh	799	858	858	858	858	856	814
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	1.01	0.966	0.966	0.966	0.966	0.960	0.927
	NPC	US\$	392,397	374,509	374,509	374,509	374,509	372,157	359,548
Power production	PV array	kWh/year	64,298	61,553	61,728	61,737	62,381	61,119	70,201
	Wind Turbine	kWh/year	11,977	11,977	11,977	11,977	11,977	11,977	11,977
	Total	kWh/year	76,275	73,530	73,706	73,715	74,358	73,096	82,179
	Unmet load	kWh/year	37.1	29.5	28.8	28.8	28.1	35.5	28.1
	Excess electricity	kWh/year	19,903	17,153	17,329	17,338	17,977	16,738	25,788
	PV penetration	%	140	134	134	134	136	133	153
	Wind penetration	%	26.0	26.0	26.0	26.0	26.0	26.0	26.0
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

3.1. PV power production

The influence of PV tracking techniques on the monthly average PV power production was examined. Six PV tracking methods were

implemented in HOMER Pro for each hub height with and without the consideration of ambient air temperature effects. The optimization results taking into account the impact of PV tracking techniques for different hub heights (17 m, 50 m, and 150 m) without and with the

Table 10

Optimization results for different PV tracking techniques for a hub height of 50 m with the consideration of the ambient air temperature effects.

Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	46.6	42.0	41.8	41.8	42.5	42.5	40.2
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	18.5	18.2	20.0	20.0	19.1	19.1	18.3
	Batteries	kWh	799	893	900	900	858	858	814
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	1.01	0.960	0.965	0.965	0.966	0.966	0.927
	NPC	US\$	392,397	372,098	374,320	374,320	374,509	374,509	359,548
Power production	PV array	kWh/year	64,298	60,901	60,741	60,750	62,381	61,591	70,201
	Wind Turbine	kWh/year	11,761	11,761	11,761	11,761	11,761	11,761	11,761
	Total	kWh/year	76,059	72,661	72,501	72,510	74,142	73,351	81,962
	Unmet load	kWh/year	38.8	36.2	38.6	38.6	34.3	27.7	33.5
	Excess electricity	kWh/year	19,675	16,277	16,118	16,127	17,756	16,972	25,565
	PV penetration	%	140	132	132	132	136	134	153
	Wind penetration	%	25.6	25.6	25.6	25.6	25.6	25.6	25.6
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

Table 11

Optimization results for different PV tracking techniques for a hub height of 150 m without taking into consideration the ambient air temperature effects.

Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	42.5	46.7	46.7	46.7	46.7	36.0	43.8
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	19.7	18.8	18.8	18.8	18.8	24.7	17.7
	Batteries	kWh	820	564	564	564	564	965	529
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	0.965	0.990	0.990	0.990	0.990	0.916	0.944
	NPC	US\$	374,033	384,054	384,054	384,054	384,054	355,281	366,111
Power production	PV array	kWh/year	58,587	67,613	67,806	67,816	68,523	52,218	76,590
	Wind Turbine	kWh/year	19,847	19,847	19,847	19,847	19,847	19,847	19,847
	Total	kWh/year	78,433	87,459	87,652	87,662	88,369	72,064	96,436
	Unmet load	kWh/year	34.0	31.7	30.7	30.5	26.6	28.4	35.6
	Excess electricity	kWh/year	22,472	31,565	31,758	31,768	32,465	16,071	40,533
	PV penetration	%	127	147	147	147	149	114	167
	Wind penetration	%	43.2	43.2	43.2	43.2	43.2	43.2	43.2
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

Table 12

Optimization results for different PV tracking techniques for a hub height of 150 m with the consideration of the ambient air temperature effects.

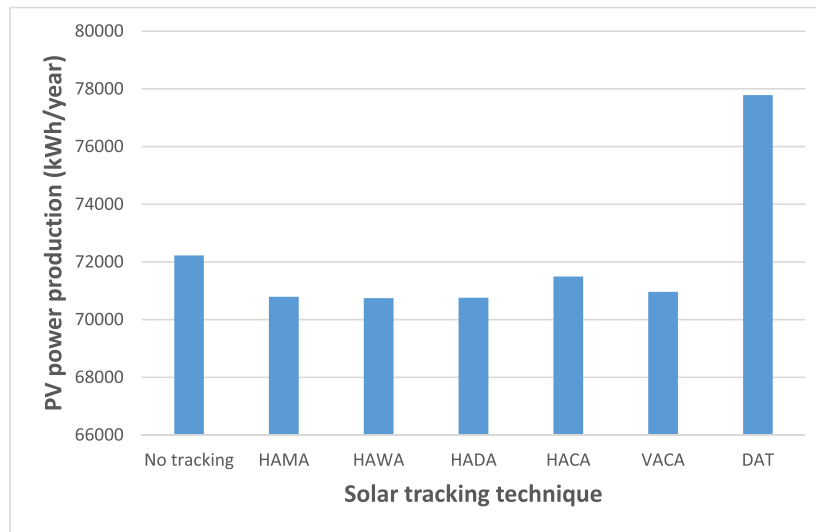
Item	Component	Unit	Best hybrid system						
			Solar tracking technique						
			No tracking	HAMA	HAWA	HADA	HACA	VACA	DAT
System architecture	PV array	kW	42.5	47.4	47.4	47.4	46.7	46.7	43.7
	Wind turbine	kW	60.0	60.0	60.0	60.0	60.0	60	60
	Converter	kW	19.7	20.1	20.1	20.1	18.8	18.8	17.6
	Batteries	kWh	820	548	548	548	564	564	538
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	CC
Cost	LCOE	US\$	0.965	1.000	1.000	1.000	0.990	0.990	0.942
	NPC	US\$	374,033	389,378	389,378	389,378	384,054	384,054	365,446
Power production	PV array	kWh/year	58,587	68,663	68,859	68,869	68,523	67,655	76,291
	Wind Turbine	kWh/year	19,486	19,486	19,486	19,486	19,486	19,486	19,486
	Total	kWh/year	78,072	88,149	88,344	88,354	88,008	87,140	95,777
	Unmet load	kWh/year	36.8	28.3	27.3	27.1	29.3	27.5	34.9
	Excess electricity	kWh/year	22,098	32,236	32,432	32,442	32,090	31,234	39,854
	PV penetration	%	127	149	150	150	149	147	166
	Wind penetration	%	42.4	42.4	42.4	42.4	42.4	42.4	42.4
Emissions	Carbon dioxide	kg/year	0	0	0	0	0	0	0

consideration of the ambient air temperature effects are presented in [Tables 7–12](#).

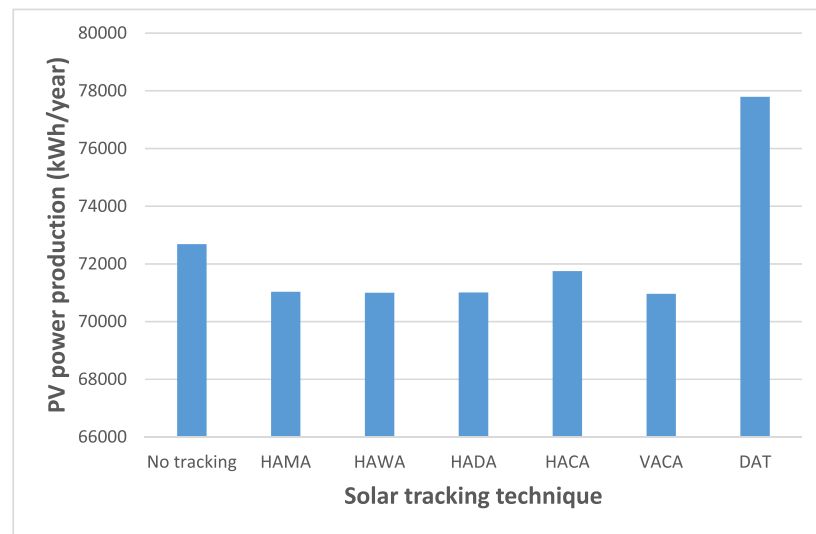
The information concerning the PV power generation were extracted from [Tables 7–12](#) and plotted in [Fig. 8](#). [Fig. 8](#) shows the PV power

production for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects ([Fig. 8a](#)), a hub height of 17 m by taking into consideration the ambient air temperature effects ([Fig. 8b](#)), a hub height of 50 m without taking

(a)



(b)



(c)

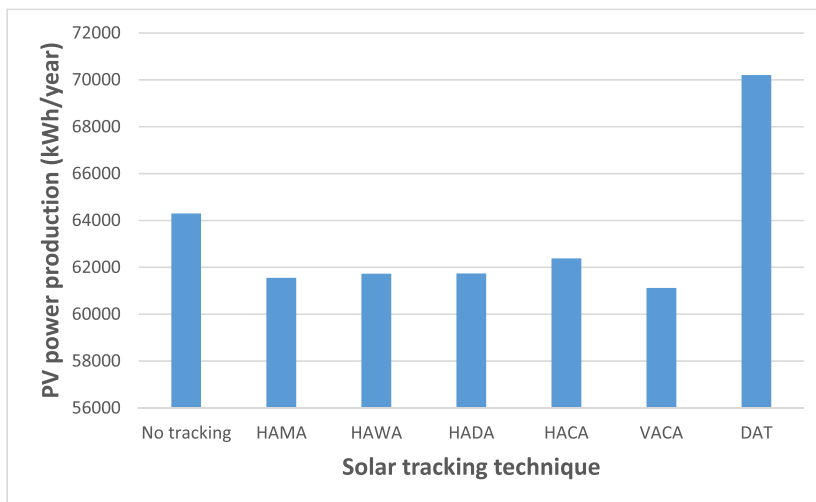
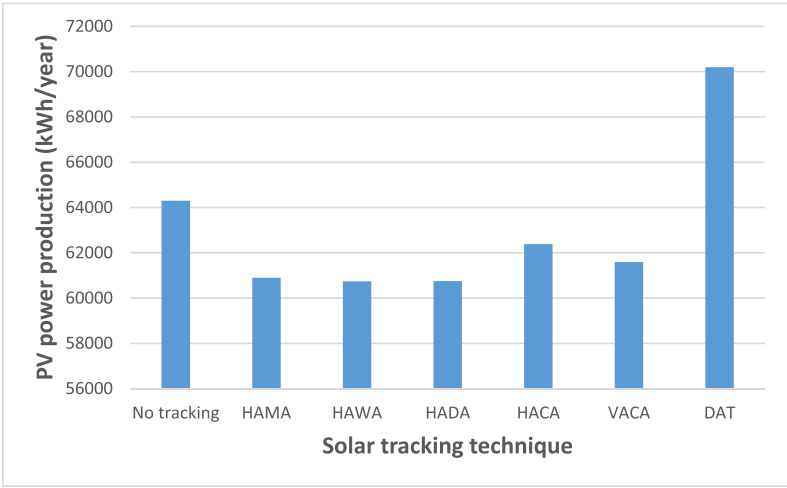
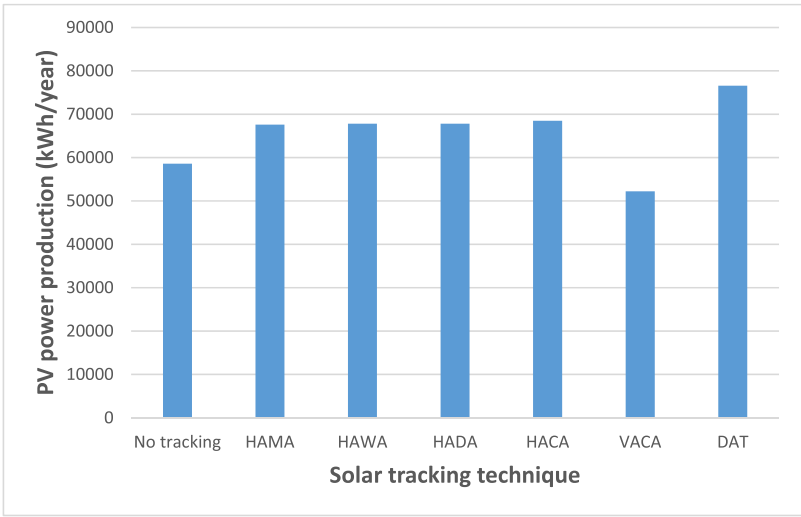


Fig. 8. PV power production for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

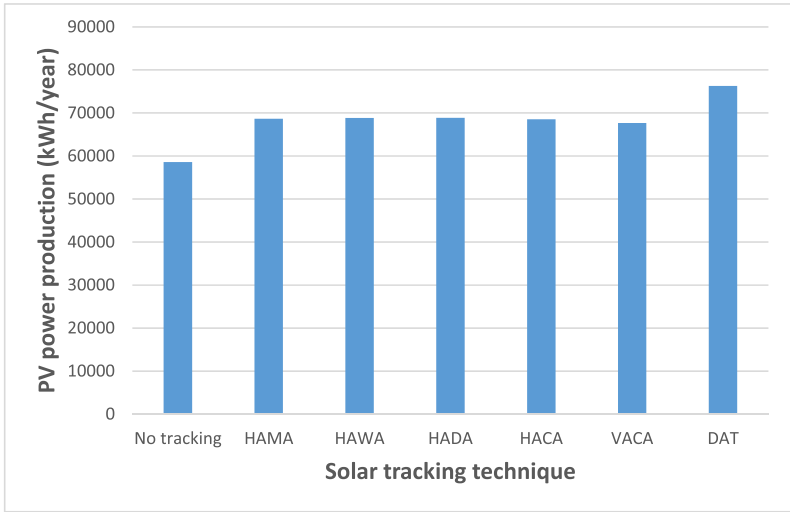


Fig. 8. (continued).

into consideration the ambient air temperature effects (Fig. 8c), a hub height of 50 m by taking into consideration the ambient air temperature effects (Fig. 8d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 8e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (Fig. 8f). It was noticed that compared to the PV power generated by

fixed solar panels, the dual-axis ST was the system which offered the maximum power production (about 7.02%–30.73% more power). This observation is in agreement with the works of the literature [28,38,50] where the authors made a similar comparison between single-axis ST and dual-axis ST techniques and concluded that the dual-axis ST offered better PV power production. In addition, from this study, it appeared

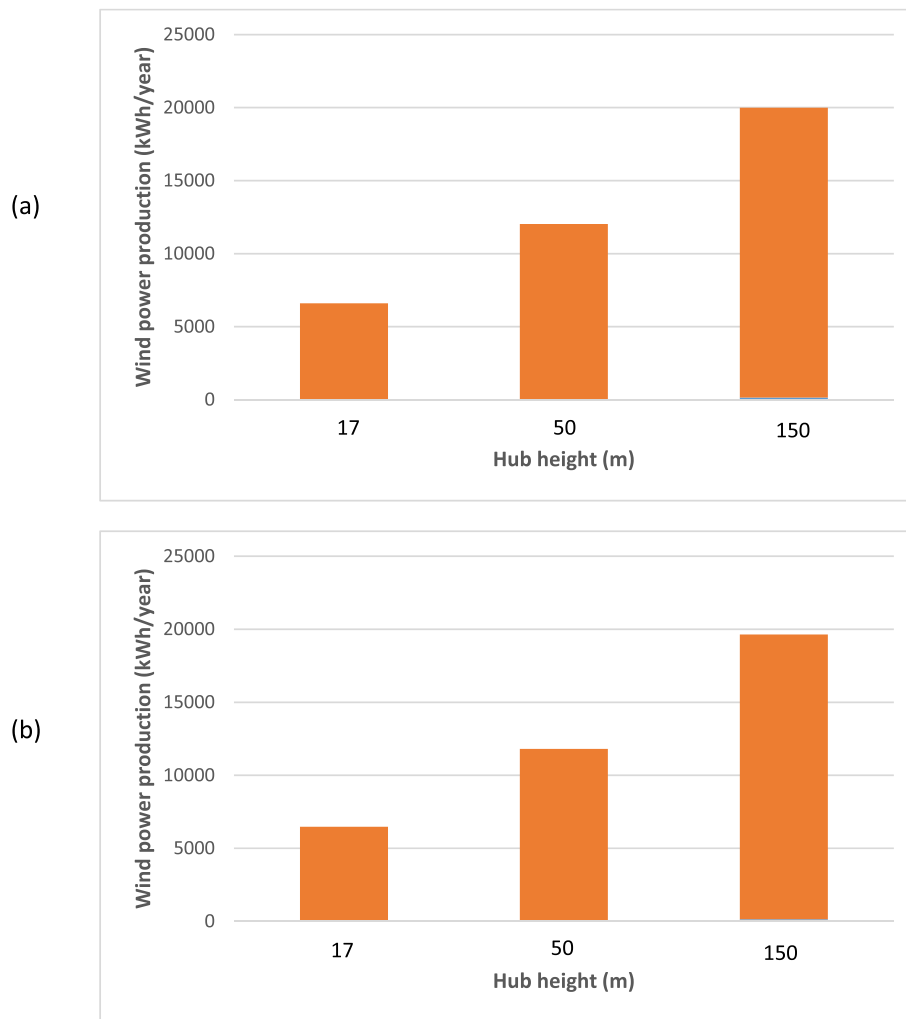


Fig. 9. Wind power production for each hub height without taking into consideration the ambient air temperature effects (a), and by taking into consideration the ambient air temperature effects (b).

that the PV power production for all ST decreased by increasing the hub height. It was also noticed that taking into account the ambient air temperature impacted the PV power production.

3.2. Wind turbine power production

The data concerning the wind turbine power output were obtained from Tables 7–12 and plotted in Fig. 9. Fig. 9 presents the wind turbine power production for each hub height without taking into consideration the ambient air temperature effects (Fig. 9a), and by taking into consideration the ambient air temperature effects (Fig. 9b). This figure allowed to conclude that the wind turbine power output significantly increased by increasing the hub height (from 82.02% for a hub height of 17 m–201.63% more power for a hub height of 150 m). It was also observed that taking into consideration the ambient air temperature slightly affected the wind turbine power production.

3.3. Net present cost

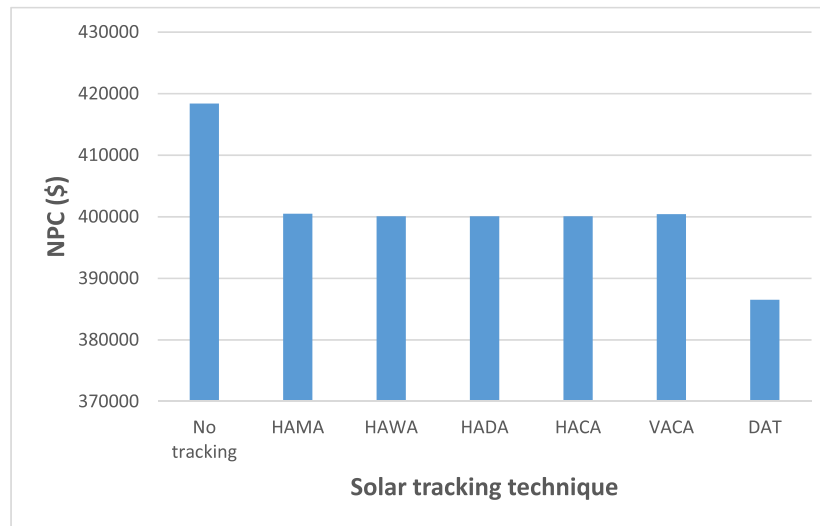
The NPC can be defined as the present value of all the costs of installing and operating of components over the project lifetime, minus the present value of all the revenues that it earns over the project lifetime. It was evaluated for all the proposed configurations and presented in Tables 7–12. It was plotted in Fig. 10 for more clarity purposes. Fig. 10 shows the NPC for each ST at: a hub height of 17 m without taking into

consideration the ambient air temperature effects (Fig. 10a), a hub height of 17 m by taking into consideration the ambient air temperature effects (Fig. 10b), a hub height of 50 m without taking into consideration the ambient air temperature effects (Fig. 10c), a hub height of 50 m by taking into consideration the ambient air temperature effects (Fig. 10d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 10e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (Fig. 10f). From these results, it was suggested that the fixed PV system had the highest NPC with an average of 395,244 \$ while the dual-axis ST system presented the lowest NPC (about 371,035 \$). This observation was not in agreement with the results of the authors in Ref. [96] where they indicated that the fixed PV system was the more profitable system in terms of NPC when compared with the dual-axis ST.

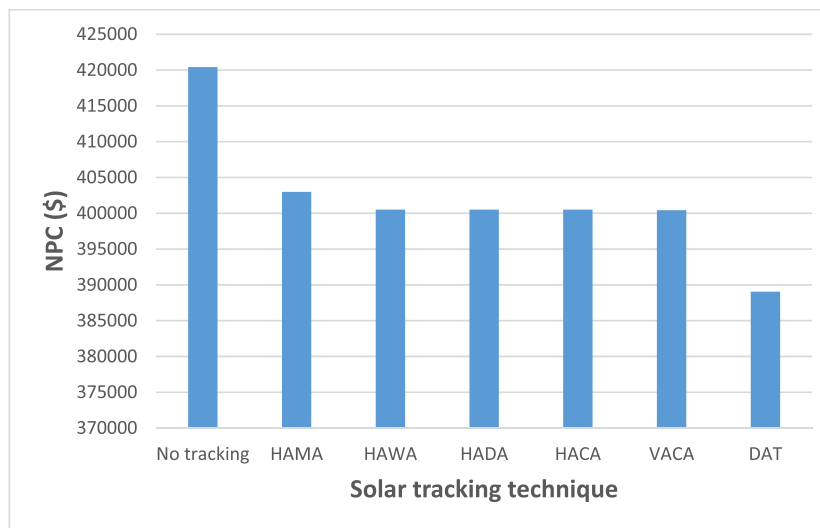
3.4. Levelized cost of energy

It is a measure of the average net present cost of electricity generation for a generator over its lifetime. It is used for investment planning and to compare different methods of electricity generation on a consistent basis. Fig. 12 depicts the LCOE for each ST at: a hub height of 17 m without taking into consideration the ambient air temperature effects (Fig. 12a), a hub height of 17 m by taking into consideration the ambient air temperature effects (Fig. 12b), a hub height of 50 m without taking into consideration the ambient air temperature effects (Fig. 12c), a hub

(a)



(b)



(c)

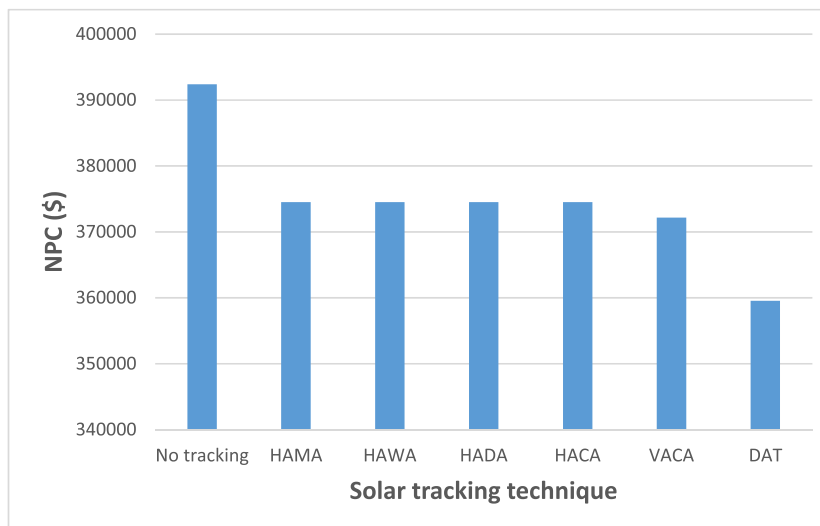
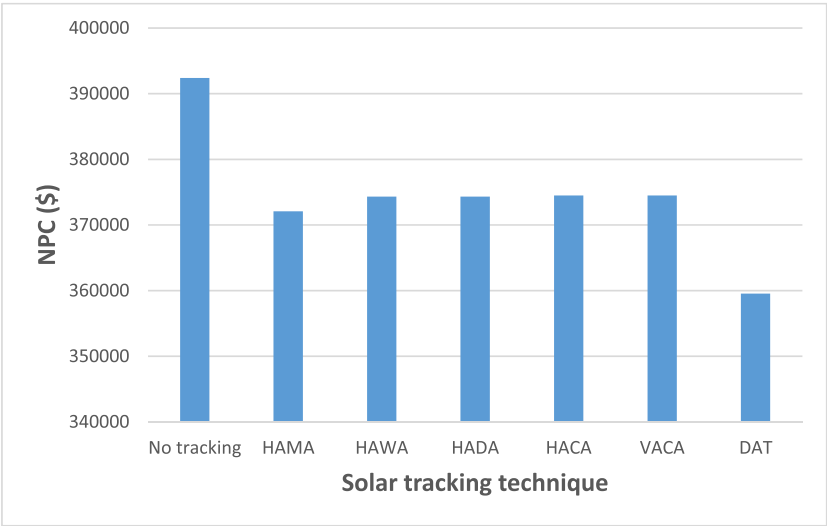
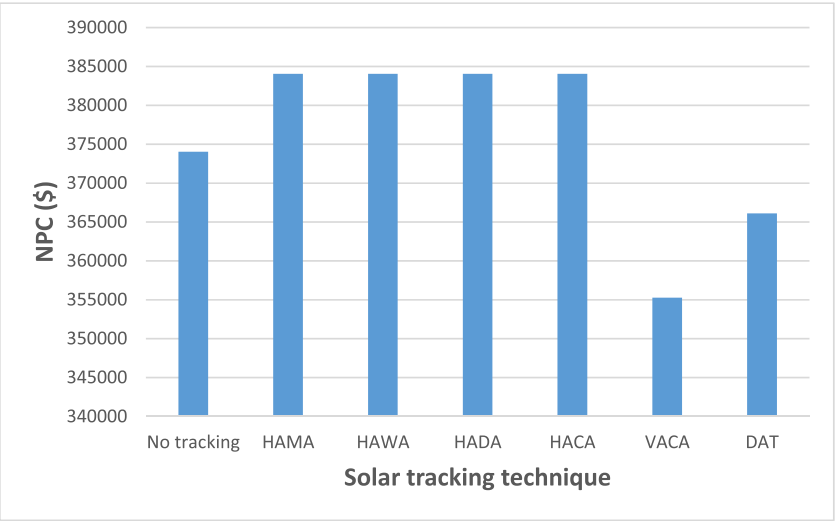


Fig. 10. NPC for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

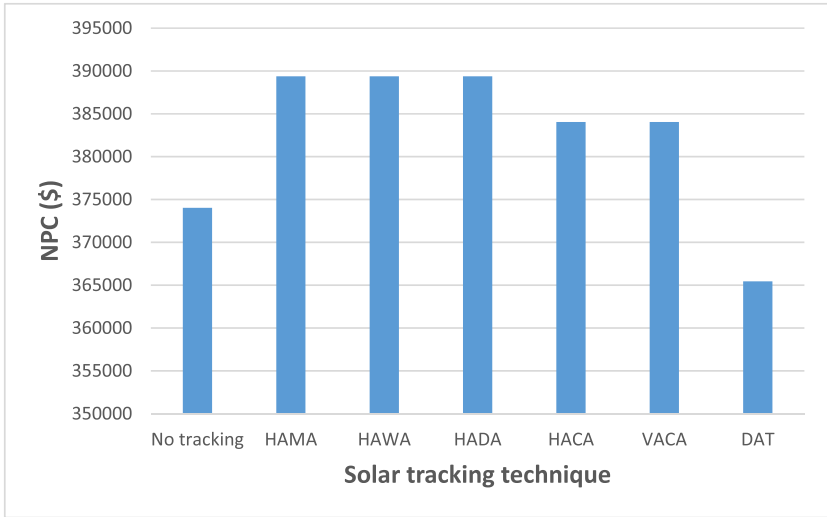
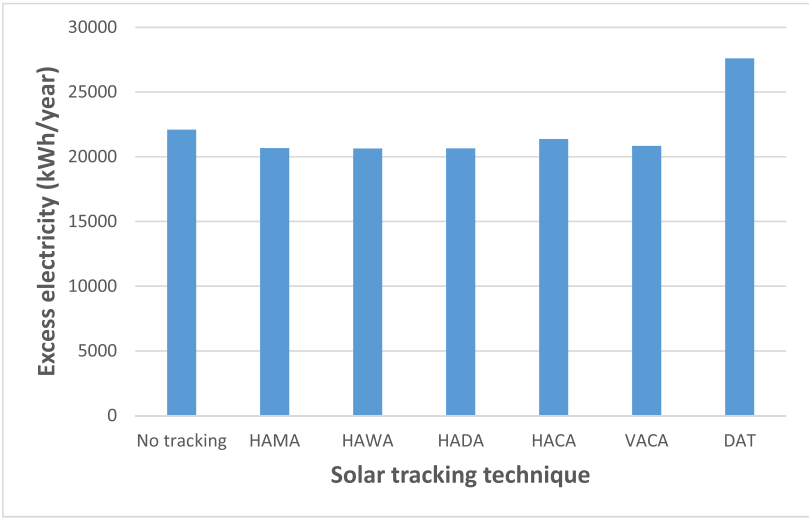


Fig. 10. (continued).

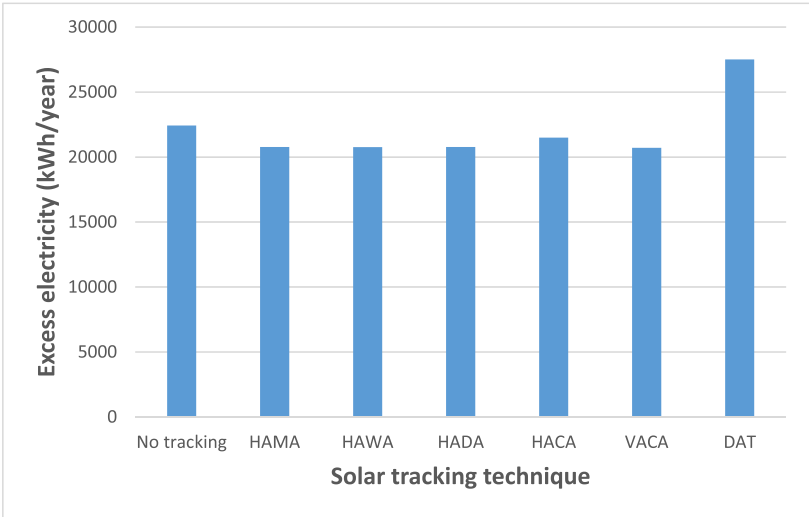
height of 50 m by taking into consideration the ambient air temperature effects (Fig. 12d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 12e), and a hub height of 150 m by taking into consideration the ambient air temperature effects

(Fig. 12f). The conclusion here followed the NPC analysis. Indeed, it was shown that the fixed PV system presented the highest average LCOE (about 1.02 \$) while the dual-axis ST system had the lowest average LCOE (about 0.956 \$).

(a)



(b)



(c)

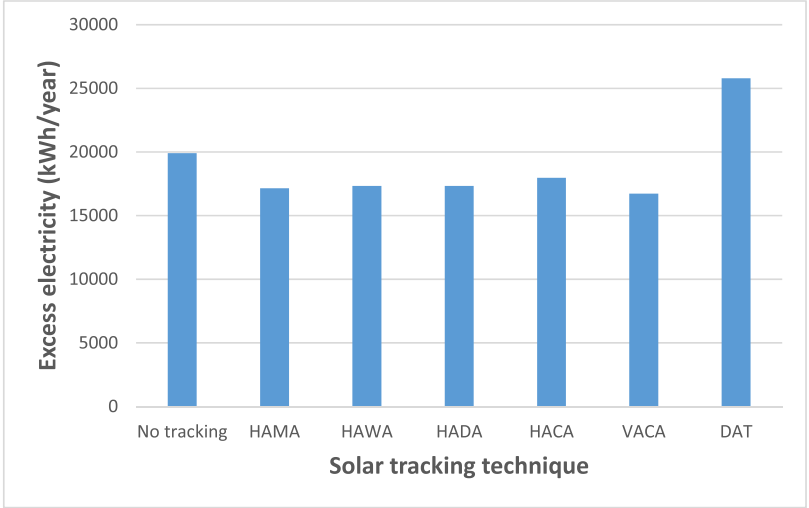
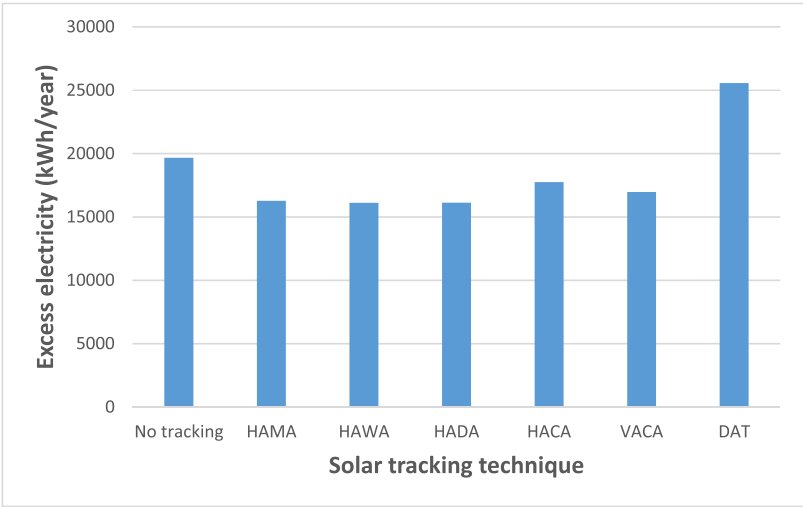
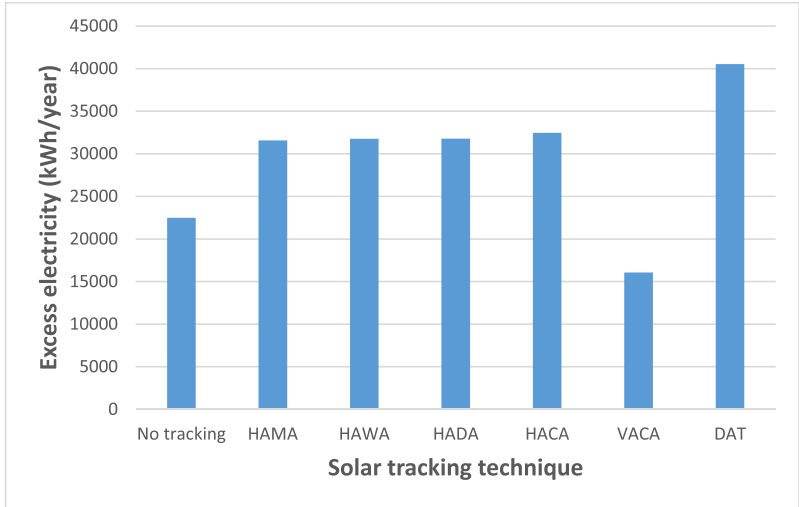


Fig. 11. Excess electricity for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

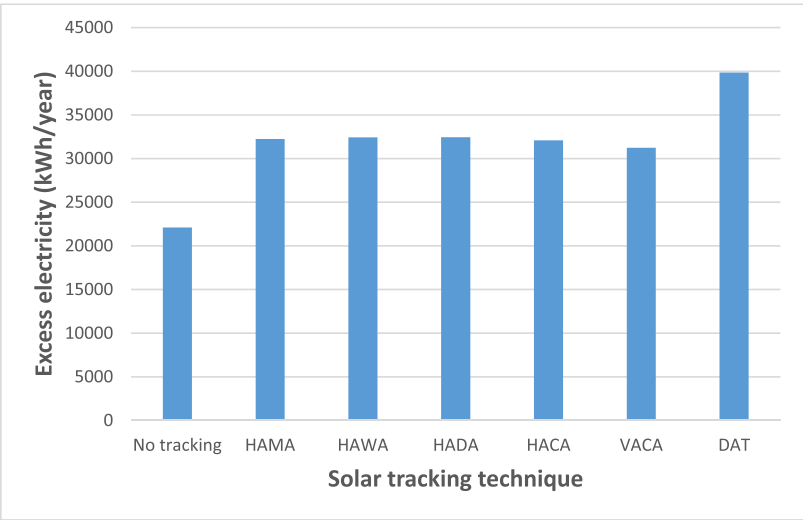


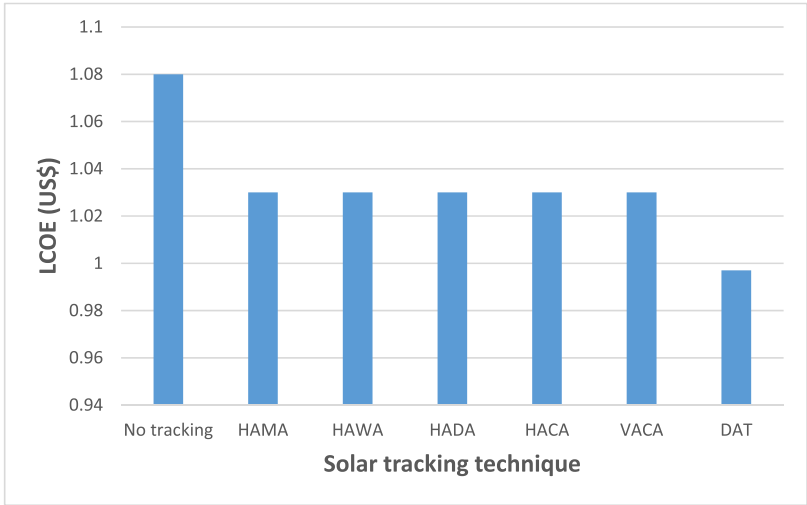
Fig. 11. (continued).

3.5. Excess electricity

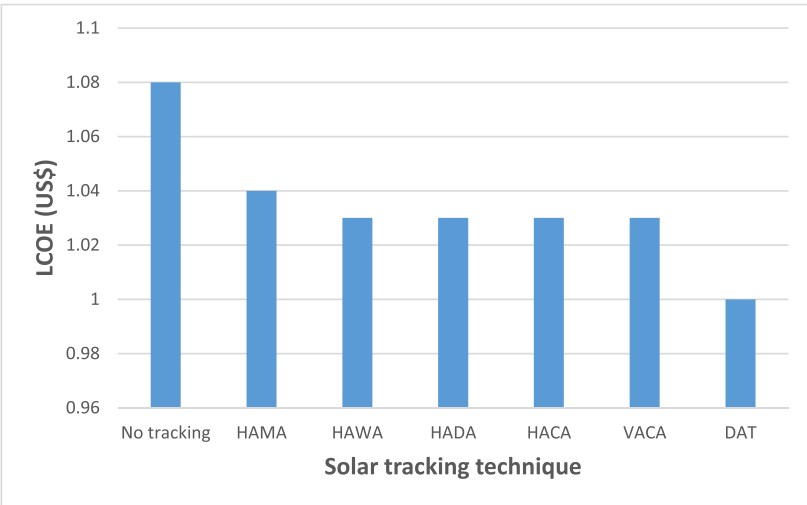
Excess electricity is surplus electrical energy that must be dumped (or curtailed) because it cannot be used to serve a load or charge batteries. The excess electricity for each ST at: a hub height of 17 m without

taking into consideration the ambient air temperature effects (Fig. 11a), a hub height of 17 m by taking into consideration the ambient air temperature effects (Fig. 11b), a hub height of 50 m without taking into consideration the ambient air temperature effects (Fig. 11c), a hub height of 50 m by taking into consideration the ambient air temperature

(a)



(b)



(c)

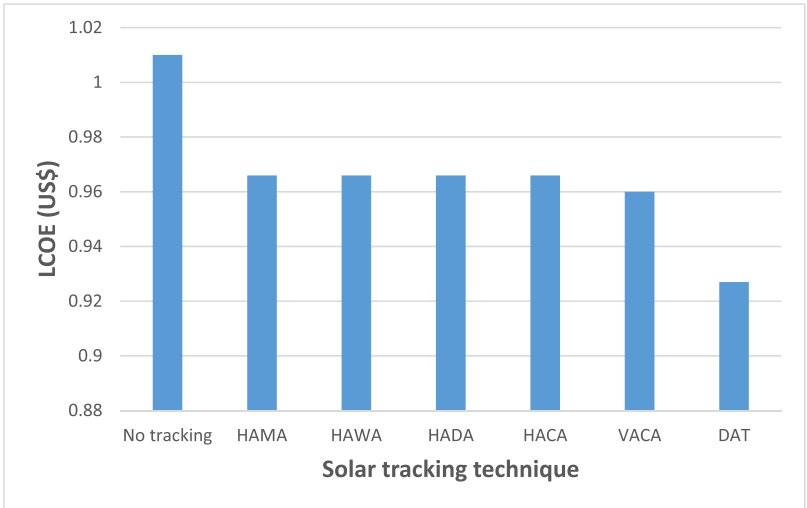
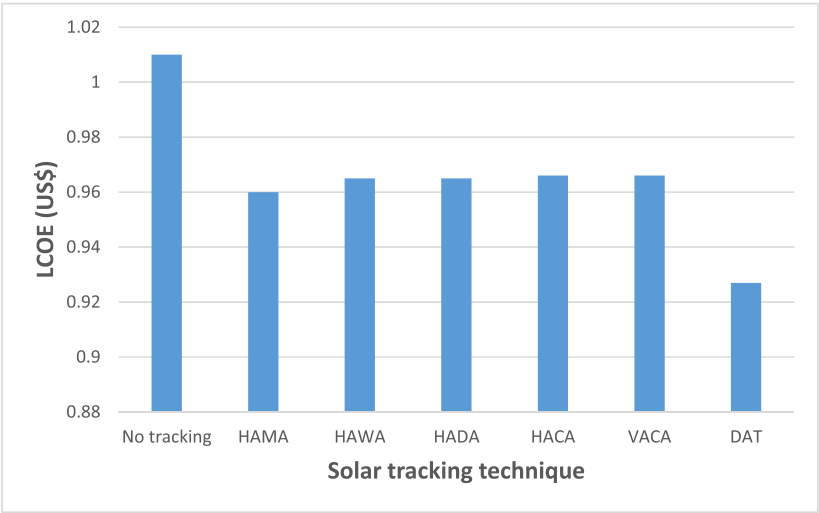
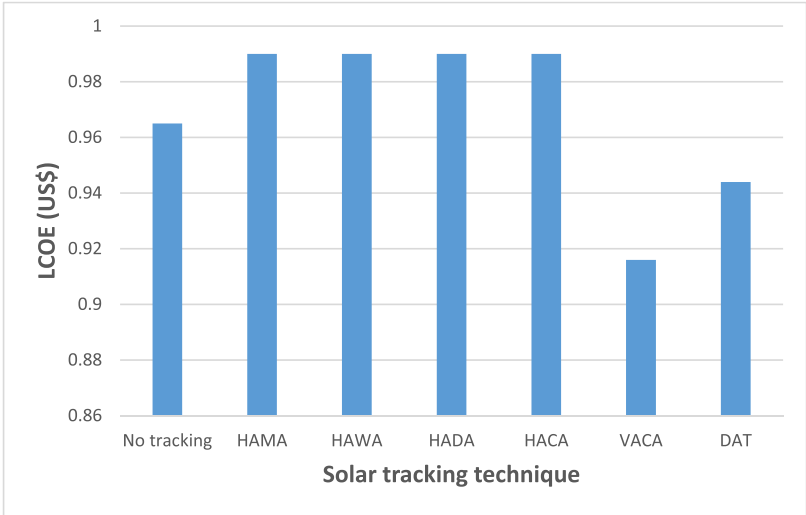


Fig. 12. LCOE for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

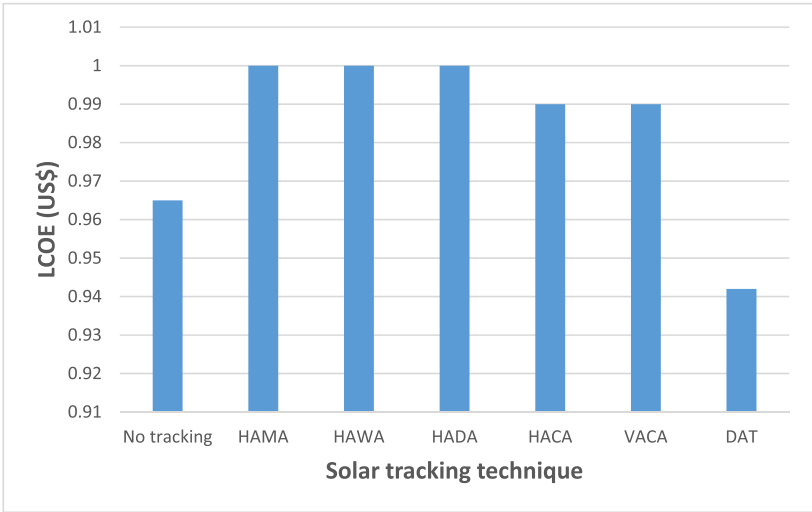


Fig. 12. (continued).

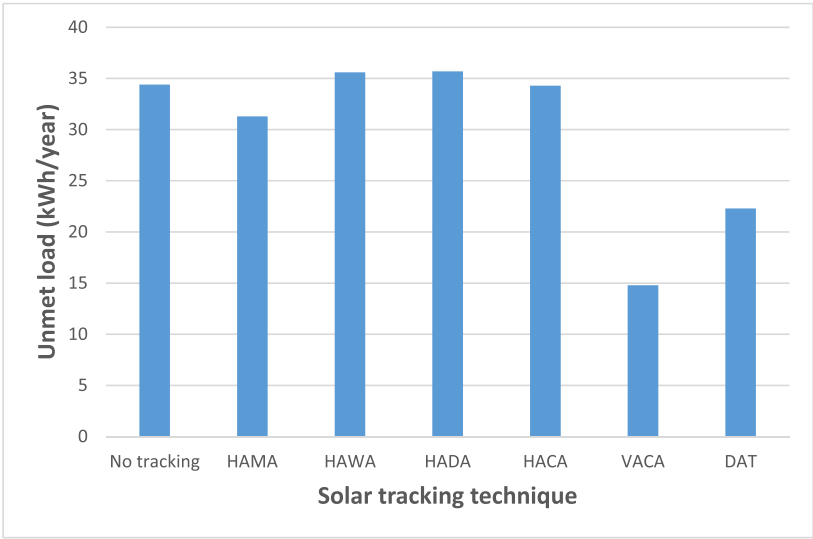
effects (Fig. 11d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 11e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (Fig. 11f) was presented. It appeared from these solutions that with the dual-axis ST, the excess electricity produced was the highest with about 31,144.667 kWh/year. This result is in accordance with the PV power

production solutions. It is worth mentioning that this excess power will be injected back to the utility for further revenues.

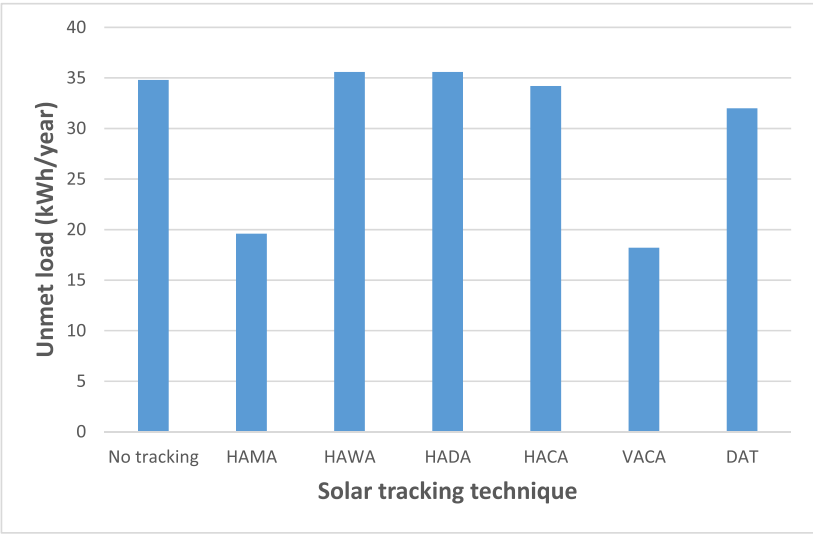
3.6. Unmet load

Unmet load is an electrical load that the power system is unable to

(a)



(b)



(c)

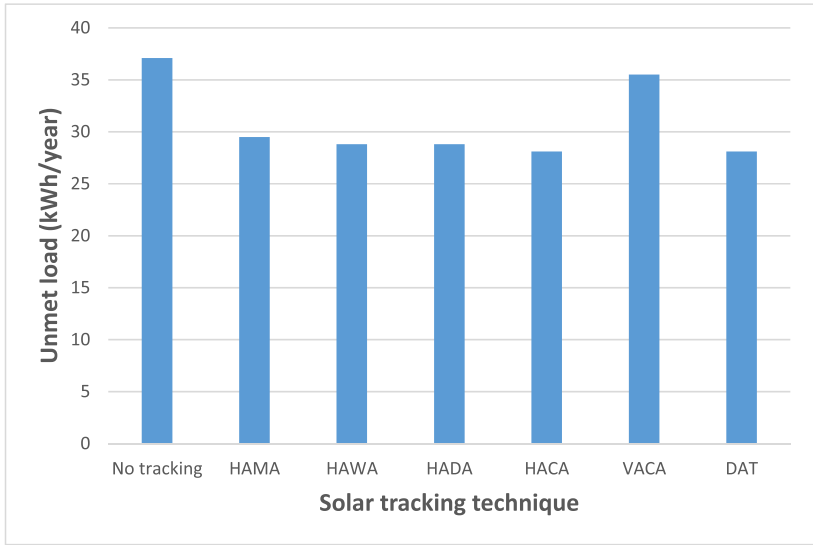
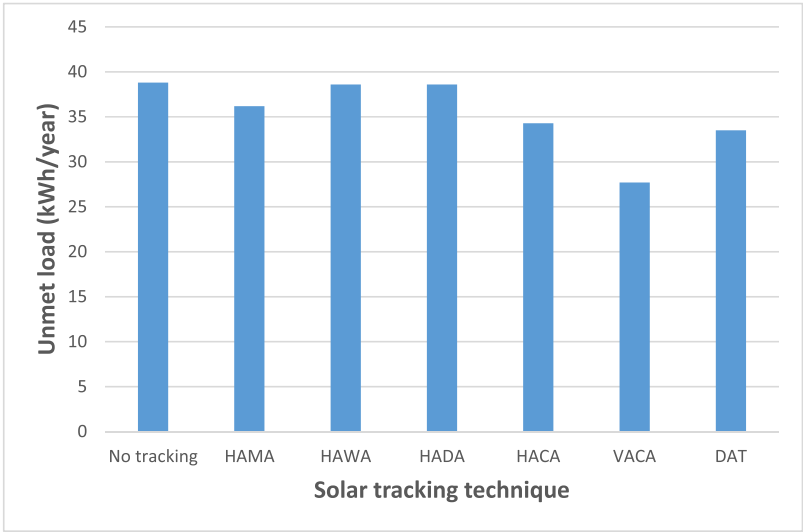
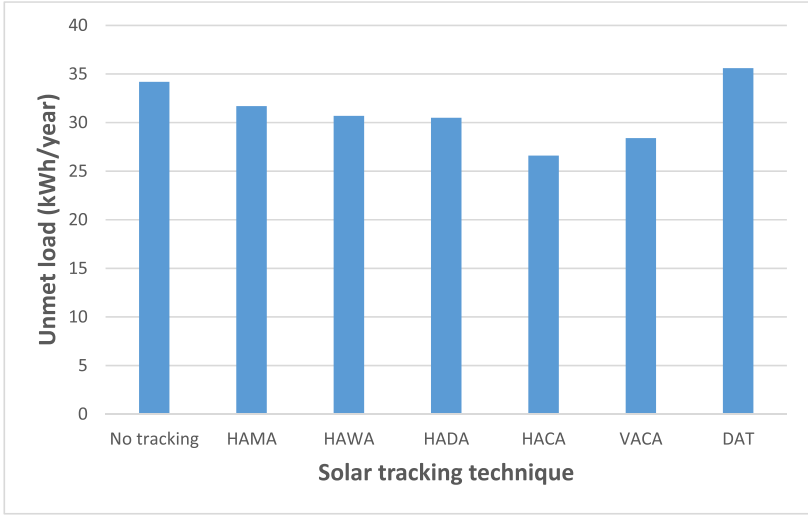


Fig. 13. Unmet load for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

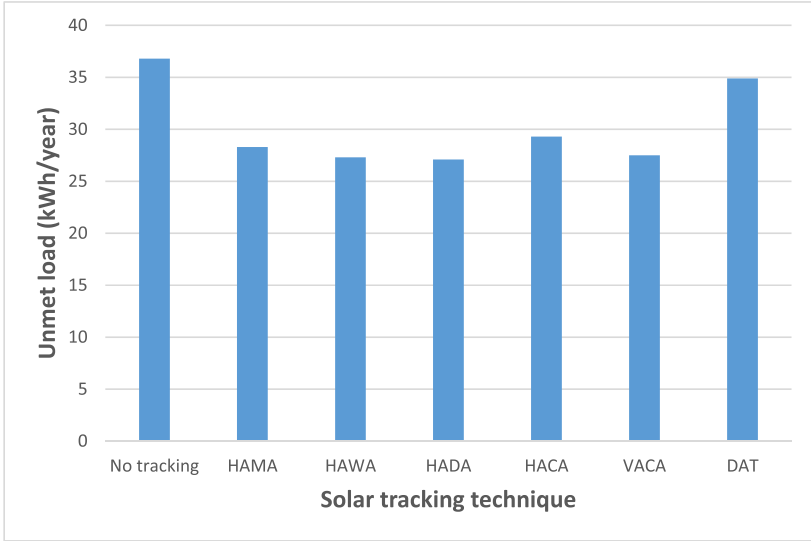
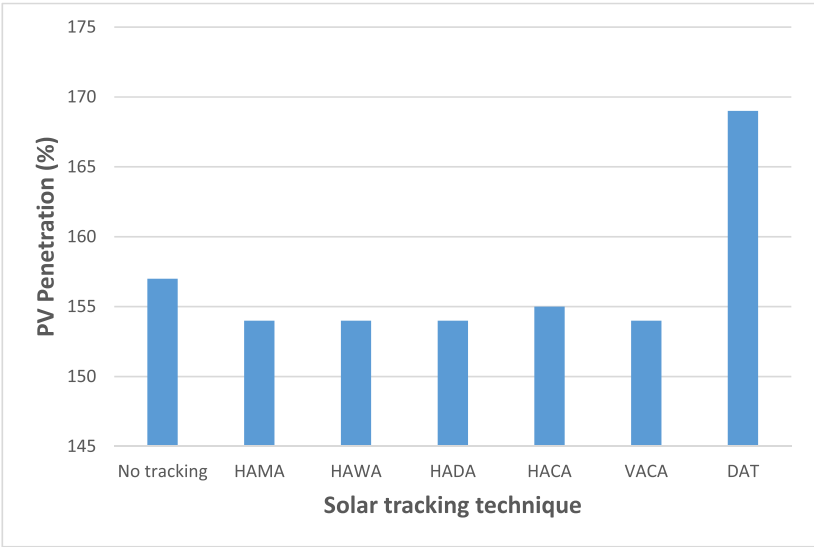


Fig. 13. (continued).

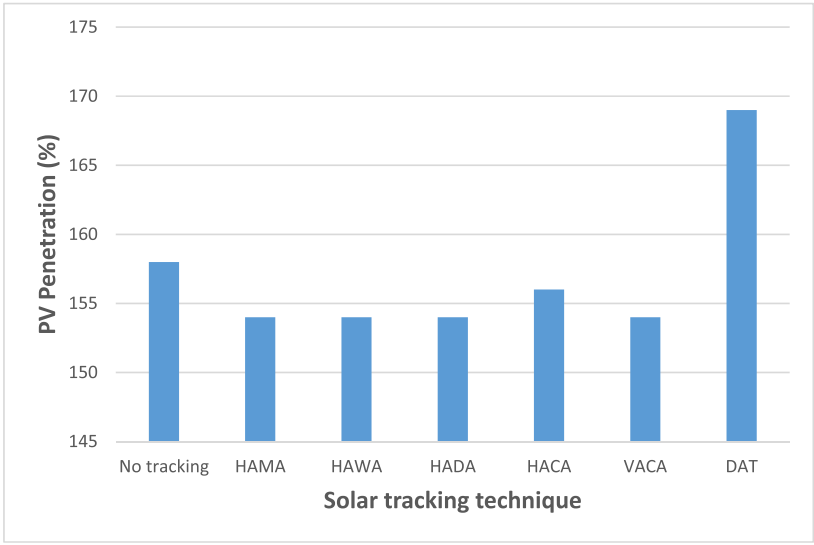
serve. It occurs when the electrical demand exceeds the supply. Fig. 11 shows the unmet load for each ST at: a hub height of 17 m without taking into consideration the ambient air temperature effects (Fig. 13a), a hub

height of 17 m by taking into consideration the ambient air temperature effects (Fig. 13b), a hub height of 50 m without taking into consideration the ambient air temperature effects (Fig. 13c), a hub height of 50 m by

(a)



(b)



(c)

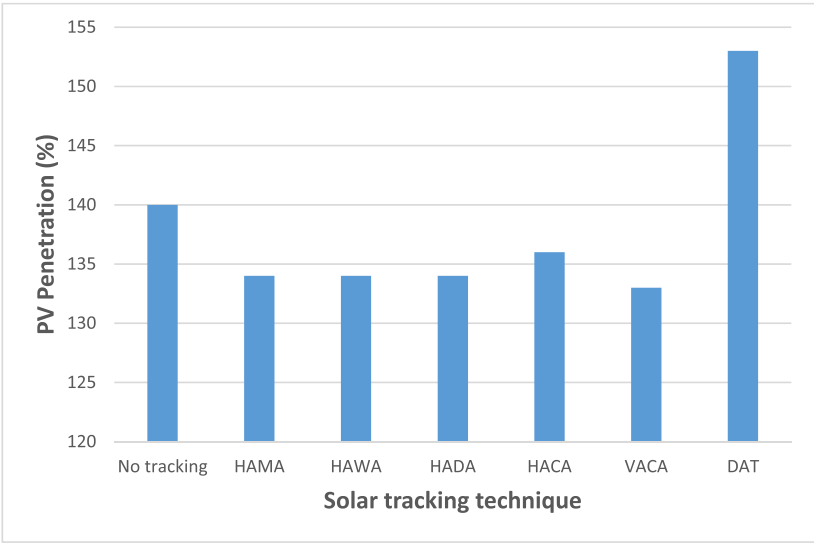
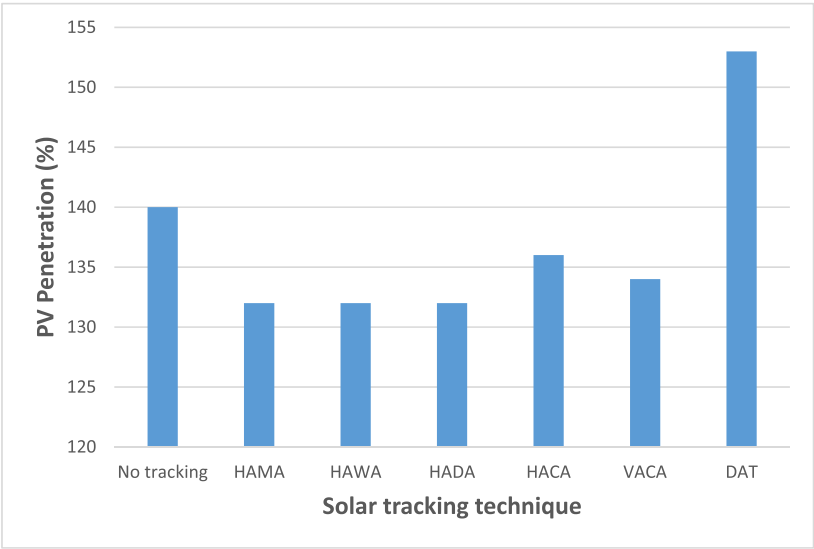
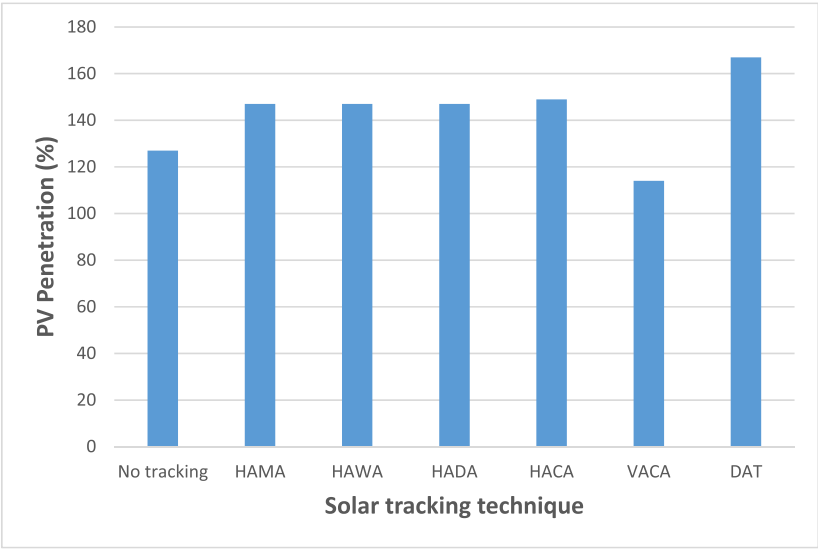


Fig. 14. PV penetration for each solar tracking technique at: a hub height of 17 m without taking into consideration the ambient air temperature effects (a), a hub height of 17 m by taking into consideration the ambient air temperature effects (b), a hub height of 50 m without taking into consideration the ambient air temperature effects (c), a hub height of 50 m by taking into consideration the ambient air temperature effects (d), a hub height of 150 m without taking into consideration the ambient air temperature effects (e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (f).

(d)



(e)



(f)

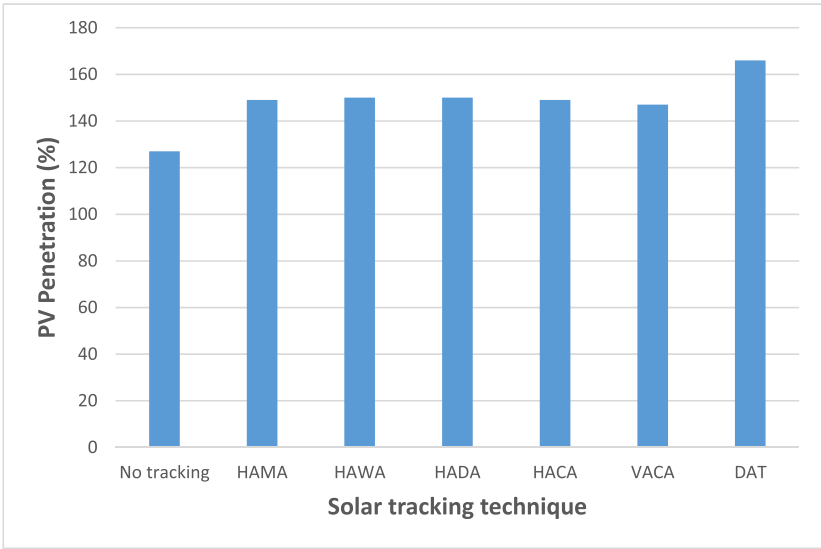


Fig. 14. (continued).

taking into consideration the ambient air temperature effects (Fig. 13d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 13e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (Fig. 13f). It was demonstrated from these figures that the vertical axis with continuous adjustment ST presented the lowest unmet load with an average of 25.35 kWh/year.

3.7. PV penetration

It is defined as the ratio of the peak PV active generation to the peak active demand of the load. Fig. 14 presents the PV penetration for each ST at: a hub height of 17 m without taking into consideration the ambient air temperature effects (Fig. 14a), a hub height of 17 m by taking into consideration the ambient air temperature effects (Fig. 14b), a hub height of 50 m without taking into consideration the ambient air temperature effects (Fig. 14c), a hub height of 50 m by taking into consideration the ambient air temperature effects (Fig. 14d), a hub height of 150 m without taking into consideration the ambient air temperature effects (Fig. 14e), and a hub height of 150 m by taking into consideration the ambient air temperature effects (Fig. 14f). The highest PV penetration was obtained here with the DAT solar tracker with an average of 162.83%. An overall observation that follows is the important

Table 13

Gas emissions for the present scenarios.

Gas emission	Value (kg/year)
Carbon dioxide	0
Carbon monoxide	0
Unburned hydrocarbons	0
Particulate matter	0
Sulfur dioxide	0
Nitrogen oxides	0

role of the PV tracking techniques for increasing PV penetration to the grid.

3.8. Wind penetration

It is the ratio of the amount of energy delivered to the power grid from wind generation to the total amount of energy delivered to the grid from all sources during a given period. Fig. 15 shows the wind penetration for each hub height without taking into consideration the ambient air temperature effects (Fig. 15a), and by taking into consideration the ambient air temperature effects (Fig. 15b). It can be simply noticed that increasing the hub height increased the wind penetration. In addition, it was concluded that the ambient air temperature slightly

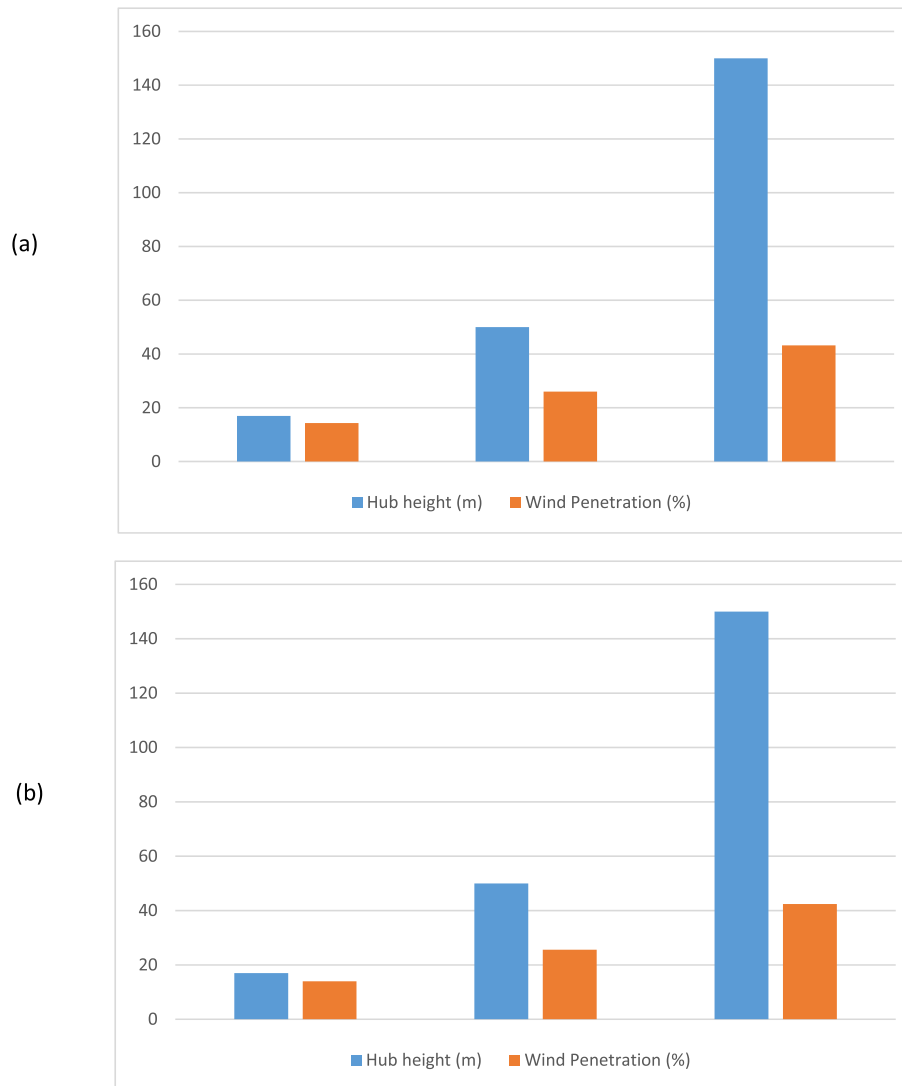


Fig. 15. Wind penetration for each hub height without taking into consideration the ambient air temperature effects (a), and by taking into consideration the ambient air temperature effects (b).

influenced the wind penetration. An average wind penetration of 27.53% was obtained in this study.

3.9. Gas emissions

HOMER pro was used in this study to estimate the gas emissions of the proposed systems. Table 13 shows the different gas emissions. It was finally concluded that no emissions of Carbon dioxide and Carbon monoxide were recorded, which clearly showed that the aforementioned systems are environmentally friendly.

4. Conclusion

The techno-economic properties of a small-scale PV/Wind/Battery hybrid system for off-grid rural electrification in the city of Mbouda were carried out. Six different ST were implemented. The impacts of ambient air temperature and hub heights were also examined. It was concluded that the dual-axis ST presented the maximum PV power production, the highest PV penetration, the highest excess of electricity, and the lowest NPC and LCOE when compared with the other solar tracker techniques. It was also noticed that the wind turbine power output significantly increased by increasing the hub height. It was demonstrated that the ambient air temperature affected PV power production. Concerning the unmet load, the vertical axis with continuous adjustment produced the lowest value. Finally, the system was environmentally friendly since no emission of Carbon dioxide and Carbon monoxide was recorded.

In future investigations, it would be interesting to integrate more power sources in the present scenarios. It would also be useful to extend the work to other towns of the country. The final results will constitute tools to support technical and economic decision-making in renewable energy investment projects in rural areas intended for political decision-makers, international investors and development partners.

Credit author statement

Pascaline TIAM KAPEN and Francis Lenine DJANNA KOFFI: Conceptualization, Gildas FOSSO TAJOUO and Pascaline TIAM KAPEN: Methodology, Pascaline TIAM KAPEN and Francis Lenine DJANNA KOFFI: Software, Pascaline TIAM KAPEN and Francis Lenine DJANNA KOFFI: Validation, Gildas FOSSO TAJOUO and Pascaline TIAM KAPEN: Investigation, Gildas FOSSO TAJOUO and Pascaline TIAM KAPEN: Writing - Original Draft, Pascaline TIAM KAPEN and Francis Lenine DJANNA KOFFI: Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] R.I. Salawu, T.A. Oduyemi, An electronic sun finder and solar tracking system, *Sol. Wind Technol.* 3 (3) (1986) 215–218.
- [2] F.M. Al-Naima, N.A. Yaghobian, Design and construction of a solar tracking system, *Sol. Wind Technol.* 7 (5) (1990) 611–617.
- [3] N.H. Helwa, A.B.G. Bahgat, A.M.R. El Shafee, El Shenawy, et al., Maximum collectable solar energy by different solar tracking systems, *Energy Sources* 22 (1) (2000) 23–34.
- [4] N.H. Helwa, A.B.G. Bahgat, A.M.R. El Shafee, El Shenawy, et al., Computation of the solar energy captured by different solar tracking systems, *Energy Sources* 22 (1) (2000) 35–44.
- [5] M.J. Clifford, D. Eastwood, Design of a novel passive solar tracker, *Sol. Energy* 77 (3) (2004) 269–280.
- [6] Z.G. Piao, J.M. Park, J.E. Kim, G.B. Cho, H.L. Baek, A study on the tracking photovoltaic system by program type, in: 2005 International Conference on Electrical Machines and Systems vol. 2, 2005, pp. 971–973.
- [7] K.S. Karimov, M.A. Saqib, P. Akhter, M.M. Ahmed, J.A. Chattha, S.A. Yousafzai, A simple photo-voltaic tracking system, *Sol. Energy Mater. Sol. Cells* 87 (1–4) (2005) 49–59.
- [8] S. Mashohor, K. Samsudin, A.M. Noor, A.R.A. Rahman, Evaluation of genetic algorithm based solar tracking system for photovoltaic panels, in: 2008 IEEE International Conference on Sustainable Energy Technologies, 2008, pp. 269–273.
- [9] N.A. Kelly, T.L. Gibson, Improved photovoltaic energy output for cloudy conditions with a solar tracking system, *Sol. Energy* 83 (11) (2009) 2092–2102.
- [10] N. Barsoum, Implementation of a prototype for a traditional solar tracking system, in: 2009 Third UKSim European Symposium on Computer Modeling and Simulation, 2009, pp. 23–30.
- [11] A. Stjepanović, S. Stjepanović, F. Softić, Z. Bundalo, Microcontroller based solar tracking system, in: 2009 9th International Conference on Telecommunication in Modern Satellite, Cable, and Broadcasting Services, 2009, pp. 518–521.
- [12] A. Argeşeanu, E. Ritchie, K. Leban, New low cost structure for dual axis mount solar tracking system using adaptive solar sensor, in: 2010 12th International Conference on Optimization of Electrical and Electronic Equipment, 2010, pp. 1109–1114.
- [13] G.M. Tina, S. Gagliano, Probabilistic modelling of hybrid solar/wind power system with solar tracking system, *Renew. Energy* 36 (6) (2011) 1719–1727.
- [14] C.S. Chin, A. Babu, W. McBride, Design, modeling and testing of a standalone single axis active solar tracker using MATLAB/Simulink, *Renew. Energy* 36 (11) (2011) 3075–3090.
- [15] X. Zhang, X. Li, K. Lu, Research on an intelligent solar tracking system based on LPC2131, in: 2012 3rd, IEEE International Conference on Network Infrastructure and Digital Content (2012) 429–432.
- [16] P. Bhattacharya, S. Mukhopadhyay, B.B. Ghosh, P.K. Bose, Optimized use of solar tracking system and wind energy, *Procedia Technology* 4 (2012) 834–839.
- [17] M. Zolkapli, S.A.M. Al-Junid, Z. Othman, A. Manut, M.M. Zulkifli, High-efficiency Dual-axis Solar Tracking Development Using Arduino, in: 2013 International Conference on Technology, Informatics, Management, Engineering and Environment, 2013, pp. 43–47.
- [18] S.B. Elagib, N.H. Osman, Design and implementation of Dual Axis solar tracker based on solar maps, in: 2013 International Conference on Computing, Electrical and Electronic Engineering (ICCEEE), 2013, pp. 697–699.
- [19] N. León, H. García, C. Ramírez, Semi-passive solar tracking concentrator, *Energy Proc.* 57 (2014) 275–284.
- [20] Y. Yao, Y. Hu, S. Gao, G. Yang, J. Du, A multipurpose dual-axis solar tracker with two tracking strategies, *Renew. Energy* 72 (2014) 88–98.
- [21] I. Stamatescu, I. Făgărăşan, G. Stamatescu, N. Arghira, S.S. Iliescu, Design and implementation of a solar-tracking algorithm, *Procedia Eng.* 69 (2014) 500–507.
- [22] M.H.M. Sidek, W.Z.W. Hasan, M.A. Kadir, S. Shafie, M.A.M. Radzi, S.A. Ahmad, M.H. Marhaban, GPS based portable dual-axis solar tracking system using astronomical equation, in: 2014 IEEE International Conference on Power and Energy (PECon), 2014, pp. 245–249.
- [23] B. Khadidja, K. Dris, A. Boubeker, S. Noureddine, Optimisation of a solar tracker system for photovoltaic power plants in Saharian region, example of Ouargla, *Energy Proc.* 50 (2014) 610–618.
- [24] H.J. Vermaak, Techno-economic analysis of solar tracking systems in South Africa, *Energy Proc.* 61 (2014) 2435–2438.
- [25] H. Bentaher, H. Kaich, N. Ayadi, M.B. Hmouda, A. Maalej, U. Lemmer, A simple tracking system to monitor solar PV panels, *Energy Convers. Manag.* 78 (2014) 872–875.
- [26] H.A. Sohag, M. Hasan, M. Khatun, M. Ahmad, An accurate and efficient solar tracking system using image processing and LDR sensor, in: 2015 2nd International Conference on Electrical Information and Communication Technologies, EICT, 2015, pp. 522–527.
- [27] Q.X. Zhang, H.Y. Yu, Q.Y. Zhang, Z.Y. Zhang, C.H. Shao, D. Yang, A solar automatic tracking system that generates power for lighting greenhouses, *Energies* 8 (7) (2015) 7367–7380.
- [28] S. Das, S. Chakraborty, P.K. Sadhu, O.S. Sastry, Design and experimental execution of a microcontroller (µC)-based smart dual-axis automatic solar tracking system, *Energy Sci. Eng.* 3 (6) (2015) 558–564.
- [29] H. Fathabadi, Novel high accurate sensorless dual-axis solar tracking system controlled by maximum power point tracking unit of photovoltaic systems, *Appl. Energy* 173 (2016) 448–459.
- [30] C.H. Huang, H.Y. Pan, K.C. Lin, Development of intelligent fuzzy controller for a two-axis solar tracking system, *Appl. Sci.* 6 (5) (2016) 130.
- [31] J. Parthipan, B.N. Raju, S. Senthilkumar, Design of one axis three position solar tracking system for paraboloidal dish solar collector, *Mater. Today: Proc.* 3 (6) (2016) 2493–2500.
- [32] S.S. Eldin, M.S. Abd-Elhady, H.A. Kandil, Feasibility of solar tracking systems for PV panels in hot and cold regions, *Renew. Energy* 85 (2016) 228–233.
- [33] M. Burhan, S.J. Oh, K.J.E. Chua, K.C. Ng, Double lens collimator solar feedback sensor and master slave configuration: development of compact and low cost two axis solar tracking system for CPV applications, *Sol. Energy* 137 (2016) 352–363.
- [34] C.C. Wei, Y.C. Song, C.C. Chang, C.B. Lin, Design of a solar tracking system using the brightest region in the sky image sensor, *Sensors* 16 (12) (2016) 1995.

- [35] Z. Mi, J. Chen, N. Chen, Y. Bai, R. Fu, H. Liu, Open-loop solar tracking strategy for high concentrating photovoltaic systems using variable tracking frequency, *Energy Convers. Manag.* 117 (2016) 142–149.
- [36] T. Hong, K. Jeong, C. Ban, J. Oh, C. Koo, J. Kim, M. Lee, A preliminary study on the 2-axis hybrid solar tracking method for the smart photovoltaic blind, *Energy Proc.* 88 (2016) 484–490.
- [37] S. Racharla, K. Rajan, Solar tracking system—a review, *Int. J. Sustain. Eng.* 10 (2) (2017) 72–81.
- [38] M.H.M. Sidek, N. Azis, W.Z.W. Hasan, M.Z.A. Ab Kadir, S. Shafie, M.A.M. Radzi, Automated positioning dual-axis solar tracking system with precision elevation and azimuth angle control, *Energy* 124 (2017) 160–170.
- [39] S.Y. Obara, K. Matsumura, S. Aizawa, H. Kobayashi, Y. Hamada, T. Suda, Development of a solar tracking system of a nonelectric power source by using a metal hydride actuator, *Sol. Energy* 158 (2017) 1016–1025.
- [40] V. Sumathi, R. Jayapragash, A. Bakshi, P.K. Akella, Solar tracking methods to maximize PV system output—A review of the methods adopted in recent decade, *Renew. Sustain. Energy Rev.* 74 (2017) 130–138.
- [41] L.M. Fernández-Ahumada, F.J. Casares, J. Ramírez-Faz, R. López-Luque, Mathematical study of the movement of solar tracking systems based on rational models, *Sol. Energy* 150 (2017) 20–29.
- [42] M. Abdollahpour, M.R. Golzarian, A. Rohani, H.A. Zarchi, Development of a machine vision dual-axis solar tracking system, *Sol. Energy* 169 (2018) 136–143.
- [43] J.A. Carballo, J. Bonilla, L. Roca, M. Berenguel, New low-cost solar tracking system based on open source hardware for educational purposes, *Sol. Energy* 174 (2018) 826–836.
- [44] A.Z. Hafez, A.M. Yousef, N.M. Harag, Solar tracking systems: technologies and trackers drive types—A review, *Renew. Sustain. Energy Rev.* 91 (2018) 754–782.
- [45] Y.M. Safan, S. Shaaban, M.I.A. El-Sebah, Performance evaluation of a multi-degree of freedom hybrid controlled dual axis solar tracking system, *Sol. Energy* 170 (2018) 576–585.
- [46] N.K. Kumar, V. Subramaniam, E. Murugan, Power analysis of non-tracking PV system with low power RTC based sensor independent solar tracking (SIST) PV system, *Mater. Today: Proc.* 5 (1) (2018) 1076–1081.
- [47] W. Nsengiyumva, S.G. Chen, L. Hu, X. Chen, Recent advancements and challenges in solar tracking systems (STS): a review, *Renew. Sustain. Energy Rev.* 81 (2018) 250–279.
- [48] M.A. Abdelghani-Idrissi, S. Khalfallaoui, D. Seguin, L. Vernières-Hassimi, S. Leveigneur, Solar tracker for enhancement of the thermal efficiency of solar water heating system, *Renew. Energy* 119 (2018) 79–94.
- [49] W. Batayneh, A. Bataineh, I. Soliman, S.A. Hafees, Investigation of a single-axis discrete solar tracking system for reduced actuations and maximum energy collection, *Autom. Constr.* 98 (2019) 102–109.
- [50] H.M. Fahad, A. Islam, M. Islam, M.F. Hasan, W.F. Brishty, M.M. Rahman, Comparative analysis of dual and single axis solar tracking system considering cloud cover, in: 2019 International Conference on Energy and Power Engineering, ICEPE, 2019, pp. 1–5.
- [51] A. Fazlizan, A. Abdulmula, A.N. Amran, C.H. Lim, K. Sopian, Performance evaluation of maximum light detection solar tracking system in the tropics, *J. Mech. Sci. Technol.* 33 (3) (2019) 1391–1397.
- [52] E.K. Mpodi, Z. Tjiparuro, O. Matsebe, Review of dual axis solar tracking and development of its functional model, *Procedia Manuf.* 35 (2019) 580–588.
- [53] K.H. Kim, S.B. Cho, An efficient concentrative photovoltaic solar system with Bayesian selection of optimal solar tracking algorithms, *Appl. Soft Comput.* 83 (2019), 105618.
- [54] A.R. Amelia, Y.M. Irwan, I. Safwati, W.Z. Leow, M.H. Mat, M.S.A. Rahim, Technologies of solar tracking systems: a review, *InOP Conference Series: Mater. Sci. Eng.* 767 (1) (2020), 012052.
- [55] M. Das, E.K. Akpinar, Investigation of the effects of solar tracking system on performance of the solar air dryer, *Renew. Energy* 167 (2021) 907–916.
- [56] O.D. Maliani, A. Bekkaoui, E.H. Baali, K. Guissi, Y. El Fellah, R. Errais, Investigation on novel design of solar still coupled with two axis solar tracking system, *Appl. Therm. Eng.* 172 (2020), 115144.
- [57] A.R. Nadia, N.A.M. Isa, M.K.M. Desa, Efficient single and dual axis solar tracking system controllers based on adaptive neural fuzzy inference system, *Journal of King Saud University-Engineering Sciences* 32 (7) (2020) 459–469.
- [58] Y. Zhu, J. Liu, X. Yang, Design and performance analysis of a solar tracking system with a novel single-axis tracking structure to maximize energy collection, *Appl. Energy* 264 (2020), 114647.
- [59] M. Das, E.K. Akpinar, Determination of thermal and drying performances of the solar air dryer with solar tracking system: apple drying test, *Case Stud. Therm. Eng.* 21 (2020), 100731.
- [60] R. ElGamal, S. Kishk, S. Al-Rejaie, G. ElMasry, Incorporation of a solar tracking system for enhancing the performance of solar air heaters in drying apple slices, *Renew. Energy* 167 (2021) 676–684.
- [61] M.A. Ismail, A. Kreshnaveyashadev, L. Ramanathan, M.H. Idris, K. Ananda-Rao, M. Mazlan, N. Fairuz, Improving the performance of solar panels by the used of dual axis solar tracking system with mirror reflection, *InJournal of Physics: Conference Series* 1432 (1) (2020), 012060.
- [62] W. Manosroi, P. Prompattra, P. Kerngburee, Performance improvement of two-axis solar tracking system by using flat-mirror reflectors, *Energy Rep.* 6 (2020) 9–14.
- [63] A. Awasthi, A.K. Shukla, M.M. SR, C. Dondariya, K.N. Shukla, D. Porwal, G. Richhariya, Review on sun tracking technology in solar PV system, *Energy Rep.* 6 (2020) 392–405.
- [64] M.A.V. Rad, A. Toopshekan, P. Rahdan, A. Kasaeian, O. Mahian, A comprehensive study of techno-economic and environmental features of different solar tracking systems for residential photovoltaic installations, *Renew. Sustain. Energy Rev.* 129 (2020), 109923.
- [65] R.F. Fuentes-Morales, A. Díaz-Ponce, M.I. Peña-Cruz, P.M. Rodrigo, L. M. Valentín-Coronado, F. Martell-Chavez, C.A. Pineda-Arellano, Control algorithms applied to active solar tracking systems: a review, *Sol. Energy* 212 (2020) 203–219.
- [66] P. Munanga, S. Chinguwa, W.R. Nyemba, C. Mbohwa, Design for manufacture and assembly of an intelligent single axis solar tracking system, *Procedia CIRP* 91 (2020) 571–576.
- [67] A. Barbón, J.A. Fernández-Rubiera, L. Martínez-Valledor, A. Pérez-Fernández, L. Bayón, Design and construction of a solar tracking system for small-scale linear Fresnel reflector with three movements, *Appl. Energy* 285 (2021), 116477.
- [68] A. Taheri, M. Malayjerdi, M. Kazemi, H. Kalani, R. Nemati-Farouji, M. Passandideh-Fard, M. Sardarabadi, Improving the performance of a nanofluid-based photovoltaic thermal module utilizing dual-axis solar tracker system: experimental examination and thermodynamic analysis, *Appl. Therm. Eng.* 196 (2021), 117178.
- [69] P. Pawar, P. Pawale, T. Nagthane, M. Thakre, N. Jangale, Performance enhancement of dual axis solar tracker system for solar panels using proteus ISIS 7.6 software package, *Global Transitions Proceedings* 2 (2) (2021) 455–460.
- [70] M. Mollahasanoglu, H.I. Okumus, Performance evaluation of the designed two-Axis solar tracking system for trabzon, *IETE J. Res.* (2021) 1–13.
- [71] G. Frydrychowicz-Jastrzębska, A. Bugala, Solar tracking system with new hybrid control in energy production optimization from photovoltaic conversion for polish climatic conditions, *Energies* 14 (10) (2021) 2938.
- [72] M. Ghassoul, A dual solar tracking system based on a light to frequency converter using a microcontroller, *Fuel Communications* 6 (2021), 100007.
- [73] M.H. Majeed, N.T. Alwan, S.E. Shcheklein, A.V. Matveev, Electromechanical solar tracker system for a parabolic dish with CPU water heater, *Mater. Today: Proc.* 42 (2021) 2346–2352.
- [74] F.Z. Baouche, B. Abderezak, A. Ladmi, K. Arbaoui, G. Cuciu, T.C. Mihaltan, F. E. Turcanu, Design and simulation of a solar tracking system for PV, *Appl. Sci.* 12 (19) (2022) 9682.
- [75] M.I. Al-Amayreh, A. Alahmer, On improving the efficiency of hybrid solar lighting and thermal system using dual-axis solar tracking system, *Energy Rep.* 8 (2022) 841–847.
- [76] C.H. Wu, H.C. Wang, H.Y. Chang, Dual-axis solar tracker with satellite compass and inclinometer for automatic positioning and tracking, *Energy for Sustainable Development* 66 (2022) 308–318.
- [77] H. Tchakounté, C.B.N. Fapi, M. Kamta, P. Wofoa, Design, experimental implementation and performance comparison of two solar tracking approaches, *Appl. Sol. Energy* 57 (1) (2021) 44–58.
- [78] H. Tchakounté, C.B.N. Fapi, M. Kamta, P. Wofoa, Experimental assessment of a smart sun tracking system consumption for the improvement of a crystalline silicon photovoltaic module performance under variable weather conditions, *Appl. Sol. Energy* 55 (6) (2019) 385–396.
- [79] C.H. Wu, H.C. Wang, H.Y. Chang, Dual-axis solar tracker with satellite compass and inclinometer for automatic positioning and tracking, *Energy for Sustainable Development* 66 (2022) 308–318.
- [80] HOMER, Energy, <https://www.homerenergy.com/>. (Accessed 27 October 2022).
- [81] Map of Cameroon, <https://reliefweb.int/map/cameroon/general-map-cameroon>. (Accessed 27 October 2022).
- [82] S. Peláez-Peláez, A. Colmenar-Santos, C. Pérez-Molina, A.E. Rosales, E. Rosales-Asensio, Techno-economic analysis of a heat and power combination system based on hybrid photovoltaic-fuel cell systems using hydrogen as an energy vector, *Energy* 224 (2021), 120110.
- [83] G. Bekele, G. Tadesse, Feasibility study of small Hydro/PV/Wind hybrid system for off-grid rural electrification in Ethiopia, *Appl. Energy* 97 (2012) 5–15.
- [84] NASA, website, <https://power.larc.nasa.gov/data-access-viewer/>. (Accessed 30 October 2022).
- [85] M.B. Kaabeche, R. Ibtiouen, Optimal sizing method for stand-alone hybrid PV/wind power generation system, *Revue des Energies Renouvelables DME* 13 (205) (2010).
- [86] <https://www.oeko-energie.de/downloads/bp350u.pdf>, 2022. (Accessed 30 October 2022).
- [87] <https://www.sunpower-uk.com/glossary/what-is-a-power-converter/>. (Accessed 1 November 2022).
- [88] M.H. Jahangir, A. Shahsavari, M.A.V. Rad, Feasibility study of a zero emission PV/Wind turbine/Wave energy converter hybrid system for stand-alone power supply: a case study, *J. Clean. Prod.* 262 (2020), 121250.
- [89] https://www.amazon.com/s?k=solar+converter&crd=POTDVSLH4M&spre fix=sconverter%2Caps%2C1092&ref=nb_sb_noss_2, 2022. (Accessed 30 October 2022).
- [90] S. Diaf, D. Daif, M. Belhamel, M. Haddidi, A. Louche, A methodology for optimal sizing of autonomous hybrid PV/wind system, *Energy Pol.* 35 (2007) 5708–5718.
- [91] <http://sunxtender.com/datasheet/PVX-2120L.pdf>, 2022. (Accessed 30 October 2022).
- [92] <http://www.windturbinewarehouse.com/pdfs/bergey/Bergey%20XL%201.0%20kW%20Turbine%20Specs.03-unenc.pdf>. (Accessed 30 October 2022).
- [93] S. Mubaarak, D. Zhang, Y. Chen, J. Liu, L. Wang, R. Yuan, M. Li, Techno-economic analysis of grid-connected pv and fuel cell hybrid system using different pv tracking techniques, *Appl. Sci.* 10 (23) (2020) 8515.
- [94] K.P. Tiam, N.B.A. Medjo, V. Chegnimonhan, G. Tchuente, R. Tchinda, Techno-economic feasibility of a PV/battery/fuel cell/electrolyzer/biogas hybrid system for energy and hydrogen production in the far north region of Cameroon by using HOMER pro, *Energy Strategy Rev.* 44 (2022), 100988.

- [95] A.I. Hassane, D.H. Didane, A.M. Tahir, R.M. Mouangue, J.G. Tamba, J. M. Hauglustaine, Comparative analysis of hybrid renewable energy systems for off-grid applications in Chad, *Int. J. Renew. Energy Dev.* 11 (1) (2022).
- [96] X. Liu, Q. Tan, Y. Niu, R. Babaei, Techno-economic analysis of solar tracker-based hybrid energy systems in a rural residential building: a case study in South Africa, *Int. J. Green Energy* (2022) 1–20.
- [97] A. Nieto, V. Vita, L. Ekonomou, N.E. Mastorakis, Economic analysis of energy storage system integration with a grid connected intermittent power plant, for power quality purposes, *Technology* 2 (5) (2016) 65–71.
- [98] A. Nieto, V. Vita, T.I. Maris, Power quality improvement in power grids with the integration of energy storage systems, *Int. J. Eng. Res. Technol.* 5 (7) (2016) 438–443.
- [99] H.M. Al Ghaithi, G.P. Fotis, V. Vita, Techno-economic assessment of hybrid energy off-grid system-A case study for Masirah island in Oman, *Int. J. Power Energy Res.* 1 (2) (2017) 103–116.
- [100] G.C. Seritan, B.A. Enache, F.C. Adochiei, F.C. Argatu, C. Christodoulou, V. Vita, F. I. Hathazi, Performance evaluation of photovoltaic panels containing cells with different bus bars configurations in partial shading conditions, *Rev. Roum. Sci. Tech.-Ser. Electrotech. Energetique.* 65 (2020) 67–70.
- [101] S. Sambhi, H. Sharma, V. Bhadoria, P. Kumar, R. Chaurasia, G. Fotis, V. Vita, Technical and economic analysis of solar PV/diesel generator smart hybrid power plant using different battery storage technologies for SRM IST, Delhi-NCR Campus, *Sustainability* 15 (4) (2023) 3666.
- [102] S. Sambhi, H. Sharma, V. Bhadoria, P. Kumar, G. Fotis, L. Ekonomou, Technical and 2E analysis of hybrid energy generating system with hydrogen production for SRM IST Delhi-NCR campus, *Design* 7 (2) (2023) 55.
- [103] R.H.T. Djiela, K.P. Tiam, G. Tchuente, Techno-economic design and performance evaluation of Photovoltaic/Diesel/Batteries system through simulation of the energy flow using generated solar radiation data, *Energy Convers. Manag.* 248 (2021), 114772.